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AUTONOMOUS MOBILE BODY, INFORMATION PROCESSING METHOD, AND INFORMATION PROCESSING DEVICE

Abstract

Provided is an information processing apparatus that includes circuitry. The circuitry recognizes at least one of a current situation or an object that is currently present around an autonomous mobile body based on at least one of sensor data or information acquired via a network, determines the recognized at least one of situation or the object currently present around the autonomous mobile body is similar to a stored situation or a stored object that is associated with a user utterance, and controls the autonomous mobile body to initiate an action associated with the user utterance based on the determination that the recognized at least one of the current situation or the object currently present around the autonomous mobile body is similar to the stored situation or the stored object that is associated with the user utterance.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a Continuation application of U.S. patent application Ser. No. 17/755,184 filed on Apr. 22, 2022, which is a U.S. National Phase of International Patent Application No. PCT/JP2020/039055 filed on Oct. 16, 2020, which claims priority benefit of Japanese Patent Application No. JP 2019-199864 filed in the Japan Patent Office on Nov. 1, 2019. Each of the above-referenced applications is hereby incorporated herein by reference in its entirety.

TECHNICAL FIELD
[0002] The present technology relates to an autonomous mobile body, an information processing method, a program, and an information processing device, and in particular, relates to an autonomous mobile body, an information processing method, a program, and an information processing device that enable a user to experience discipline for the autonomous mobile body.
BACKGROUND ART
[0003] Attempts have conventionally been made to bring actions of animal-type autonomous mobile bodies closer to actions of actual animals. For example, an animal-type autonomous mobile body has been devised that performs marking behavior for indicating territory (see, for example, Patent Document 1).
CITATION LIST
Patent Document
[0004] Patent Document 1: WO 2019/138618 A
SUMMARY OF THE INVENTION
Problems to be Solved by the Invention
[0005] By the way, one of the pleasures of keeping a pet is to discipline the pet so that it can perform a desired action. Thus, by making it possible to discipline an autonomous mobile body similarly to an actual pet, it can be expected that the user experience for the autonomous mobile body will be improved.
[0006] The present technology has been made in view of such a situation, and enables a user to experience discipline for the autonomous mobile body.
Solutions to Problems
[0007] An autonomous mobile body of one aspect of the present technology includes: a recognition unit that recognizes an instruction given; an action planning unit that plans an action on the basis of the instruction recognized; and an operation control unit that controls execution of the action planned, in which the action planning unit changes a detail of a predetermined action as an action instruction that is an instruction for the predetermined action is repeated.
[0008] An information processing method of one aspect of the present technology: recognizes an instruction given to an autonomous mobile body; plans an action of the autonomous mobile body on the basis of the instruction recognized, and changes a detail of a predetermined action as an action instruction that is an instruction for the predetermined action is repeated; and controls execution of the action of the autonomous mobile body planned.

[0009] A program of one aspect of the present technology causes a computer to execute processing of: recognizing an instruction given to an autonomous mobile body; planning an action of the autonomous mobile body on the basis of the instruction recognized, and changing a detail of a predetermined action as an action instruction that is an instruction for the predetermined action is repeated; and controlling execution of the action of the autonomous mobile body planned.
[0010] An information processing device of one aspect of the present technology includes: a recognition unit that recognizes an instruction given to an autonomous mobile body; an action planning unit that plans an action of the autonomous mobile body on the basis of the instruction recognized; and an operation control unit that controls execution of the action of the autonomous mobile body planned, in which the action planning unit changes a detail of a predetermined action by the autonomous mobile body as an action instruction that is an instruction for the predetermined action is repeated.
[0011] According to one aspect of the present technology, the instruction given to the autonomous mobile body is recognized, the action of the autonomous mobile body is planned on the basis of the instruction recognized, and the detail of the predetermined action is changed as the action instruction that is the instruction for the predetermined action is repeated, and the action of the autonomous mobile body planned is executed.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0012] FIG. **1** is a block diagram illustrating an embodiment of an information processing system to which the present technology is applied.
[0013] FIG. **2** is a diagram illustrating a hardware configuration example of an autonomous mobile body.
[0014] FIG. **3** is a configuration example of an actuator included in the autonomous mobile body.
[0015] FIGS. **4**A, **4**B, and **4**C are diagrams for explaining a function of a display included in the autonomous mobile body.
[0016] FIGS. **5**A and **5**B are diagrams illustrating an operation example of the autonomous mobile body.
[0017] FIG. **6** is a block diagram illustrating a functional configuration example of the autonomous mobile body.
[0018] FIG. **7** is a block diagram illustrating a configuration example of an expected value calculation unit of the autonomous mobile body.
[0019] FIG. **8** is a block diagram illustrating a functional configuration example of an information processing terminal.
[0020] FIG. **9** is a block diagram illustrating a functional configuration example of an information processing server.
[0021] FIGS. **10**A and **10**B are diagrams illustrating an outline of pattern recognition learning based on teaching.
[0022] FIGS. **11**A and **11**B are diagrams illustrating an outline of motion recognition learning based on teaching.
[0023] FIGS. **12**A and **12**B are diagrams illustrating an outline of spatial area recognition learning based on teaching.
[0024] FIGS. **13**A and **13**B are diagrams for explaining teaching using a marker.
[0025] FIG. **14** is a diagram for explaining teaching using wireless communication.
[0026] FIGS. **15**A and **15**B are diagrams for explaining teaching using an inaudible sound.
[0027] FIG. **16** is a flowchart for explaining pattern recognition learning processing executed by the autonomous mobile body.

[0028] FIG. **17** is a flowchart for explaining a first embodiment of action control processing executed by the autonomous mobile body.
[0029] FIG. **18** is a graph illustrating an example of time series transitions of a promotion expected value, a suppression expected value, and a habit expected value.
[0030] FIG. **19** is a graph illustrating an example of time series transitions of total expected values over time.
[0031] FIG. **20** is a diagram for explaining a method of calculating the habit expected value.
[0032] FIG. **21** is a flowchart for explaining a second embodiment of the action control processing executed by the autonomous mobile body.
[0033] FIG. **22** is a flowchart for explaining details of action place learning processing.
[0034] FIG. **23** is a diagram illustrating an example of clustering of indicated places.
[0035] FIG. **24** is a graph illustrating an example of a time series transition of a urinary intention coefficient over time.
[0036] FIG. **25** is a flowchart for explaining a third embodiment of the action control processing executed by the autonomous mobile body.
[0037] FIG. **26** is a diagram illustrating an example of a user interface of the information processing terminal.
[0038] FIG. **27** is a diagram illustrating an example of the user interface of the information processing terminal.
[0039] FIG. **28** is a diagram illustrating an example of the user interface of the information processing terminal.
[0040] FIG. **29** is a diagram illustrating an example of the user interface of the information processing terminal.
[0041] FIG. **30** is a diagram illustrating an example of the user interface of the information processing terminal.
[0042] FIG. **31** is a diagram illustrating an example of the user interface of the information processing terminal.
[0043] FIG. **32** is a diagram illustrating an example of the user interface of the information processing terminal.
[0044] FIG. **33** is a diagram illustrating an example of the user interface of the information processing terminal.
[0045] FIG. **34** is a diagram illustrating an example of the user interface of the information processing terminal.
[0046] FIG. **35** is a diagram illustrating an example of the user interface of the information processing terminal.
[0047] FIG. **36** is a diagram illustrating a configuration example of a computer.

MODE FOR CARRYING OUT THE INVENTION

[0048] The following is a description of embodiments for carrying out the present technology. The description will be given in the following order. [0049] 1. Embodiment [0050] 2. Modifications
[0051] 3. Others

1. Embodiment

[0052] First, an embodiment of the present technology will be described with reference to FIGS. **1**, **2**, **3**, **4**A, **4**B, **4**C, 5A, **5**B, **6**, **7**, **8**, **9**, **10**A, **10**B, **11**A, **11**B, **12**A, **12**B, **13**A, **13**B, **14**, **15**A, **15**B, **16**, **17**, **18**, **19**, **20**, **21**, **22**, **23**, **24**, **25**, **26**, **27**, **28**, **29**, **30**, **31**, **32**, **33**, **34**, and **35**.

<Configuration Example of Information Processing System **1**>

[0053] FIG. **1** is a block diagram illustrating an embodiment of an information processing system **1** to which the present technology is applied.
[0054] The information processing system **1** includes an autonomous mobile body **11-1** to an autonomous mobile body **11**-*n*, an information processing terminal **12-1** to an information processing terminal **12**-*n*, and an information processing server **13**.

[0055] Note that, hereinafter, in a case where it is not necessary to individually distinguish the autonomous mobile body **11-1** to the autonomous mobile body **11**-*n* from each other, it is simply referred to as an autonomous mobile body **11**. Hereinafter, in a case where it is not necessary to individually distinguish the information processing terminal **12-1** to the information processing terminal **12**-*n* from each other, it is simply referred to as an information processing terminal **12**.
[0056] Between each autonomous mobile body **11** and the information processing server **13**, between each information processing terminal **12** and the information processing server **13**, between each autonomous mobile body **11** and each information processing terminal **12**, between the autonomous mobile bodies **11**, and between the information processing terminals **12**, communication is possible via the network **21**. Furthermore, it is also possible to directly communicate between each autonomous mobile body **11** and each information processing terminal **12**, between the autonomous mobile bodies **11**, and between the information processing terminals **12**, without going through the network **21**.
[0057] The autonomous mobile body **11** is an information processing device that recognizes situations of the autonomous mobile body **11** and its surroundings on the basis of collected sensor data and the like, and autonomously selects and executes various operations depending on the situations. One of features of the autonomous mobile body **11** is that it autonomously executes an appropriate operation depending on a situation, unlike a robot that simply performs an operation depending on a user's instruction.
[0058] For example, the autonomous mobile body **11** can execute user recognition, object recognition, and the like based on a captured image, and can perform various autonomous actions depending on a recognized user, object, and the like. Furthermore, the autonomous mobile body **11** can also execute, for example, voice recognition based on the user's utterance and perform an action based on the user's instruction or the like.
[0059] Moreover, the autonomous mobile body **11** performs pattern recognition learning to acquire abilities of the user recognition and the object recognition. At this time, the autonomous mobile body **11** can perform not only supervised learning based on given learning data but also perform pattern recognition learning related to the object or the like by dynamically collecting learning data on the basis of teaching by the user or the like.
[0060] Furthermore, the autonomous mobile body **11** can be disciplined by the user. Here, discipline for the autonomous mobile body **11** is broader than general discipline in which rules and prohibitions are taught and caused to be remembered, for example, and means that a change appears that can be felt by the user in the autonomous mobile body **11** due to that the user interacts the autonomous mobile body **11**.
[0061] A shape, an ability, and a level of desire or the like of the autonomous mobile body **11** can be appropriately designed depending on a purpose and a role. For example, the autonomous mobile body **11** is configured by an autonomous mobile robot that autonomously moves in a space and executes various operations. Specifically, for example, the autonomous mobile body **11** is configured by an autonomous mobile robot having a shape and operational ability imitating an animal such as a human or a dog. Furthermore, for example, the autonomous mobile body **11** is configured by a vehicle or another device having an ability to communicate with the user.
[0062] The information processing terminal **12** includes, for example, a smartphone, a tablet terminal, a personal computer (PC), or the like, and is used by the user of the autonomous mobile body **11**. The information processing terminal **12** implements various functions by executing a predetermined application program (hereinafter, simply referred to as an application). For example, the information processing terminal **12** communicates with the information processing server **13** via the network **21** or directly communicates with the autonomous mobile body **11**, to collect various data regarding the autonomous mobile body **11** and present the data to the user, or give an instruction to the autonomous mobile body **11**.
[0063] For example, the information processing server **13** collects various data from each

autonomous mobile body **11** and each information processing terminal **12**, provides various data to each autonomous mobile body **11** and each information processing terminal **12**, or controls operation of the autonomous mobile body **11**. Furthermore, for example, the information processing server **13** can also perform pattern recognition learning, and processing corresponding to the user's discipline, similarly to the autonomous mobile body **11**, on the basis of the data collected from each autonomous mobile body **11** and each information processing terminal **12**. Moreover, for example, the information processing server **13** supplies various data regarding the above-described application and each autonomous mobile body **11** to each information processing terminal **12**.

[0064] The network **21** includes some of, for example, public line networks such as the Internet, a telephone line network, and a satellite communication network, various local area networks (LANs) including Ethernet (registered trademark), a wide area network (WAN), and the like. Furthermore, the network **21** may include a dedicated line network such as an Internet protocol-virtual private network (IP-VPN). Furthermore, the network **21** may include a wireless communication network such as Wi-Fi (registered trademark) or Bluetooth (registered trademark).

[0065] Note that, the configuration of the information processing system **1** can be flexibly changed depending on specifications, operation, and the like. For example, the autonomous mobile body **11** may further perform information communication with various external devices in addition to the information processing terminal **12** and the information processing server **13**. The external devices may include, for example, a server that transmits weather, news, and other service information, and various home appliances owned by the user.

[0066] Furthermore, for example, the autonomous mobile body **11** and the information processing terminal **12** do not necessarily have to have a one-to-one relationship, and may have, for example, a many-to-many, many-to-one, or one-to-many relationship. For example, one user can confirm data regarding a plurality of the autonomous mobile bodies **11** by using one information processing terminal **12**, or confirm data regarding one autonomous mobile body **11** by using a plurality of information processing terminals.

<Hardware Configuration Example of Autonomous Mobile Body **11**>

[0067] Next, a hardware configuration example of the autonomous mobile body **11** will be described. Note that, in the following, a case where the autonomous mobile body **11** is a dog-shaped quadruped walking robot will be described as an example.

[0068] FIG. **2** is a diagram illustrating the hardware configuration example of the autonomous mobile body **11**. The autonomous mobile body **11** is the dog-shaped quadruped walking robot including a head, a body, four legs, and a tail.

[0069] The autonomous mobile body **11** includes two displays, a display **51**L and a display **51**R, on the head. Note that, hereinafter, in a case where it is not necessary to individually distinguish the display **51**L and the display **51**R from each other, it is simply referred to as displays **51**.

[0070] Furthermore, the autonomous mobile body **11** includes various sensors. The autonomous mobile body **11** includes, for example, a microphone **52**, a camera **53**, a Time Of Flight (ToF) sensor **525**, a motion sensor **55**, a distance measurement sensor **56**, a touch sensor **57**, an illuminance sensor **58**, sole buttons **59**, and inertial sensors **60**.

[0071] The autonomous mobile body **11** includes, for example, four microphones **52** on the head. Each microphone **52** collects ambient sounds including, for example, the user's utterances and ambient environmental sounds. Furthermore, by including a plurality of microphones **52**, it is possible to collect sounds generated in the surroundings with high sensitivity and to localize a sound source.

[0072] The autonomous mobile body **11** includes, for example, two wide-angle cameras **53** at the tip of a nose and a waist, and captures images of the surroundings of the autonomous mobile body **11**. For example, the camera **53** arranged at the tip of the nose captures an image in a front field of view (that is, a dog's field of view) of the autonomous mobile body **11**. The camera **53** arranged on the waist captures an image of the surroundings centered on the upper part of the autonomous mobile body **11**. For example, the autonomous mobile body **11** can implement Simultaneous Localization and Mapping (SLAM) by extracting feature points on the ceiling, or the like, on the basis of the image captured by the camera **53** arranged on the waist.

[0073] The ToF sensor **54** is provided, for example, at the tip of the nose and detects a distance to an object existing in front of the head. The autonomous mobile body **11** can detect distances to various objects with high accuracy by the ToF sensor **54**, and can implement an operation depending on a relative position with a target object including the user, an obstacle, or the like.

[0074] The motion sensor **55** is arranged on a chest, for example, and detects a location of the user or a pet kept by the user. The autonomous mobile body **11** can implement various operations with respect to a moving body, for example, operations corresponding to emotions such as interest, fear, and surprise, by detecting the moving body existing in front by the motion sensor **55**.

[0075] The distance measurement sensor **56** is arranged on the chest, for example, and detects a situation of a floor surface in front of the autonomous mobile body **11**. The autonomous mobile body **11** can detect a distance to an object existing on the floor surface in front with high accuracy by the distance measurement sensor **56**, and can implement an operation depending on a relative position with the object.

[0076] The touch sensor **57** is arranged at a portion where the user is highly likely to touch of the autonomous mobile body **11**, for example, the top of the head, under the chin, the back, or the like, and detects contact by the user. The touch sensor **57** is configured by, for example, a capacitance type or pressure sensitive type touch sensor. The autonomous mobile body **11** can detect a contact action such as touching, stroking, hitting, or pushing by the user by the touch sensor **57**, and can perform an operation depending on the contact action.

[0077] The illuminance sensor **58** is arranged at the base of the tail on the back surface of the head, for example, and detects an illuminance in the space where the autonomous mobile body **11** is located. The autonomous mobile body **11** can detect ambient brightness by the illuminance sensor **58** and execute an operation depending on the brightness.

[0078] The sole buttons **59** are, for example, arranged at portions corresponding to paws of the four legs, respectively, and detect whether or not bottom surfaces of the legs of the autonomous mobile body **11** are in contact with the floor. The autonomous mobile body **11** can detect contact or non-contact with the floor surface by the sole buttons **59**, and can grasp, for example, that the autonomous mobile body **11** is lifted up by the user.

[0079] The inertial sensors **60** are arranged, for example, on the head and the body, respectively, and detects physical quantities such as velocity, acceleration, and rotation of the head and the body. For example, each of the inertial sensors **60** is configured by a 6-axis sensor that detects acceleration and angular velocity on the X axis, Y axis, and Z axis. The autonomous mobile body **11** can detect motions of the head and the body with high accuracy by the inertial sensors **60**, and can implement operation control depending on a situation.

[0080] Note that, the configuration of the sensor included in the autonomous mobile body **11** can be flexibly changed depending on the specifications, operation, and the like. For example, in addition to the configuration described above, the autonomous mobile body **11** may further include, for example, a temperature sensor, a geomagnetic sensor, various communication devices including a Global Navigation Satellite System (GNSS) signal receiver, and the like.

[0081] Next, a configuration example of joint units of the autonomous mobile body **11** will be described with reference to FIG. **3**. FIG. **3** illustrates a configuration example of actuators **71** included in the autonomous mobile body **11**. The autonomous mobile body **11** has a total of 22 degrees of freedom of rotation, which are rotation points illustrated in FIG. **3** and additionally two each in the ear and the tail and one in the mouth.

[0082] For example, the autonomous mobile body **11** has three degrees of freedom in the head, so that it can achieve both nodding and tilting of the neck. Furthermore, the autonomous mobile body

11 can implement a natural and flexible motion closer to that of a real dog by reproducing the swing motion of the waist by the actuators **71** provided in the waist.
[0083] Note that, the autonomous mobile body **11** may implement the 22 degrees of freedom of rotation described above by combining, for example, a 1-axis actuator and a 2-axis actuator. For example, the 1-axis actuator may be adopted for elbow and knee portions of the leg, and 2-axis actuator may be adopted for the shoulder and the base of the thigh.
[0084] Next, a function of the displays **51** included in the autonomous mobile body **11** will be described with reference to FIGS. **4**A, **4**B, and **4**C.
[0085] The autonomous mobile body **11** includes the two displays **51**R and **51**L corresponding to the right eye and the left eye, respectively. The displays **51** each have a function of visually expressing eye motions and emotions of the autonomous mobile body **11**. For example, the displays **51** can express motions of eyeballs, pupils, and eyelids depending on emotions and motions to produce natural motions similar to those of real animals such as dogs, and to express a line-of-sight and emotions of the autonomous mobile body **11** with high accuracy and flexibility. Furthermore, the user can intuitively grasp a state of the autonomous mobile body **11** from the motions of the eyeballs displayed on the displays **51**.
[0086] Furthermore, the displays **51** are implemented by, for example, two independent Organic Light Emitting Diodes (OLEDs). By using the OLEDs, it is possible to reproduce curved surfaces of the eyeballs. As a result, a more natural exterior can be implemented as compared with a case where one flat display is used to express a pair of eyeballs and a case where two independent flat displays are used to express two eyeballs.
[0087] With the configuration described above, the autonomous mobile body **11** reproduces motions and emotional expressions closer to those of a real organism by controlling motions of the joint units and eyeballs with high accuracy and flexibility, as illustrated in FIGS. **5**A and **5**B.
[0088] Note that, FIGS. **5**A and **5**B are diagrams illustrating an operation example of the autonomous mobile body **11**; however, in FIGS. **5**A and **5**B, since a description will be given by focusing on the motions of the joint units and eyeballs of the autonomous mobile body **11**, an external structure of the autonomous mobile body **11** is illustrated in a simplified manner.
<Functional Configuration Example of Autonomous Mobile Body **11**>
[0089] Next, a functional configuration example of the autonomous mobile body **11** will be described with reference to FIG. **6**. The autonomous mobile body **11** includes an input unit **101**, a communication unit **102**, an information processing unit **103**, a drive unit **104**, an output unit **105**, and a storage unit **106**.
[0090] The input unit **101** includes various sensors and the like illustrated in FIG. **2**, and has a function of collecting various sensor data regarding situations of the user and the surroundings. Furthermore, the input unit **101** includes, for example, input devices such as a switch and a button. The input unit **101** supplies the collected sensor data and input data input via the input devices to the information processing unit **103**.
[0091] The communication unit **102** communicates with another autonomous mobile body **11**, the information processing terminal **12**, and the information processing server **13** via the network **21** or without the network **21**, and transmits and receives various data. The communication unit **102** supplies the received data to the information processing unit **103**, and acquires the data to be transmitted from the information processing unit **103**.
[0092] Note that, a communication method of the communication unit **102** is not particularly limited, and can be flexibly changed depending on the specifications and operation.
[0093] The information processing unit **103** includes, for example, a processor such as a Central Processing Unit (CPU), performs various types of information processing, and controls each unit of the autonomous mobile body **11**. The information processing unit **103** includes a recognition unit **121**, a learning unit **122**, an action planning unit **123**, and an operation control unit **124**.
[0094] The recognition unit **121** recognizes a situation in which the autonomous mobile body **11** is

placed on the basis of the sensor data and input data supplied from the input unit **101** and the received data supplied from the communication unit **102**. The situation in which the autonomous mobile body **11** is placed includes, for example, the situations of the autonomous mobile body **11** and the surroundings. The situation of the autonomous mobile body **11** includes, for example, the state and motion of the autonomous mobile body **11**. The situations of the surroundings include, for example, a state, motion, and instruction of a surrounding person such as the user, a state and motion of a surrounding organism such as a pet, a state and motion of a surrounding object, a time, a place, a surrounding environment, and the like. The surrounding object includes, for example, another autonomous mobile body. Furthermore, to recognize the situation, the recognition unit **121** performs, for example, person identification, recognition of a facial expression or line-of-sight, emotion recognition, object recognition, motion recognition, spatial area recognition, color recognition, shape recognition, marker recognition, obstacle recognition, step recognition, brightness recognition, temperature recognition, voice recognition, word understanding, position estimation, posture estimation, and the like.

[0095] Furthermore, the recognition unit **121** has a function of estimating and understanding the situation on the basis of various types of recognized information. At this time, the recognition unit **121** may comprehensively estimate the situation by using knowledge stored in advance.

[0096] The recognition unit **121** supplies data indicating a result of recognition or a result of estimation of the situation (hereinafter referred to as situation data) to the learning unit **122** and the action planning unit **123**. Furthermore, the recognition unit **121** registers the data indicating the result of recognition or the result of estimation of the situation in action history data stored in the storage unit **106**.

[0097] The action history data is data indicating a history of the action of the autonomous mobile body **11**. The action history data includes items of, for example, the date and time when an action is started, the date and time when the action is completed, a trigger for executing the action, a place where an instruction for the action is given (however, in a case where the place is indicated), a situation when the action is performed, and whether or not the action is completed (whether or not the action is executed to the end).

[0098] As the trigger for executing the action, for example, in a case where the action is executed triggered by the user's instruction, a detail of the instruction is registered. Furthermore, for example, in a case where the action is executed triggered by a predetermined situation, a detail of the situation is registered. Moreover, for example, in a case where the action is executed triggered by an object indicated by the user or a recognized object, a type of the object is registered.

[0099] The learning unit **122** learns the situation and the action, and an influence of the action on the environment, on the basis of the sensor data and input data supplied from the input unit **101**, the received data supplied from the communication unit **102**, the situation data supplied from the recognition unit **121**, data regarding the action of the autonomous mobile body **11** supplied from the action planning unit **123**, and the action history data stored in the storage unit **106**. For example, the learning unit **122** performs the pattern recognition learning described above, or performs learning of an action pattern corresponding to the user's discipline.

[0100] For example, the learning unit **122** implements the learning described above by using a machine learning algorithm such as deep learning. Note that, a learning algorithm adopted by the learning unit **122** is not limited to the example described above, and can be appropriately designed.

[0101] The learning unit **122** supplies data indicating a learning result (hereinafter referred to as learning result data) to the action planning unit **123** or stores the data in the storage unit **106**.

[0102] The action planning unit **123** plans an action to be performed by the autonomous mobile body **11** on the basis of the recognized or estimated situation and the learning result data. The action planning unit **123** supplies data indicating the planned action (hereinafter referred to as action plan data) to the operation control unit **124**. Furthermore, the action planning unit **123** supplies data regarding the action of the autonomous mobile body **11** to the learning unit **122**, or registers the data in the action history data stored in the storage unit **106**.

[0103] The operation control unit **124** controls the operation of the autonomous mobile body **11** to execute the planned action by controlling the drive unit **104** and the output unit **105** on the basis of the action plan data. The operation control unit **124** performs, for example, rotation control of the actuators **71**, display control of the displays **51**, voice output control with the speaker, and the like on the basis of the action plan.

[0104] The drive unit **104** bends and stretches the plurality of joint units of the autonomous mobile body **11** on the basis of the control by the operation control unit **124**. More specifically, the drive unit **104** drives the actuators **71** included in the respective joint units on the basis of the control by the operation control unit **124**.

[0105] The output unit **105** includes, for example, the displays **51**, the speaker, a haptics device, and the like, and outputs visual information, auditory information, tactile information, and the like on the basis of the control by the operation control unit **124**.

[0106] The storage unit **106** includes, for example, a non-volatile memory and a volatile memory, and stores various programs and data.

<Configuration Example of Expected Value Calculation Unit **151**>

[0107] FIG. **7** illustrates a configuration example of an expected value calculation unit **151** that is a part of functions implemented by the learning unit **122** and the action planning unit **123**.

[0108] The expected value calculation unit **151** calculates an expected value indicating a degree of expectation of the user for the autonomous mobile body **11** to perform a predetermined action. The expected value calculation unit **151** includes a promotion expected value calculation unit **161**, a suppression expected value calculation unit **162**, a habit expected value calculation unit **163**, and a total expected value calculation unit **164**.

[0109] The promotion expected value calculation unit **161** calculates the promotion expected value on the basis of the sensor data and input data supplied from the input unit **101** and the received data supplied from the communication unit **102**. The promotion expected value indicates the user's short-term expected value for the autonomous mobile body **11** to execute a predetermined action. For example, the promotion expected value changes when the user gives an instruction to promote the predetermined action. The promotion expected value calculation unit **161** supplies data indicating the calculated promotion expected value to the total expected value calculation unit **164**.

[0110] The suppression expected value calculation unit **162** calculates the suppression expected value on the basis of the sensor data and input data supplied from the input unit **101** and the received data supplied from the communication unit **102**. The suppression expected value indicates the user's short-term expected value for the autonomous mobile body **11** to suppress a predetermined action. For example, the suppression expected value changes when the user gives an instruction to suppress the predetermined action. The suppression expected value calculation unit **162** supplies data indicating the calculated suppression expected value to the total expected value calculation unit **164**.

[0111] The habit expected value calculation unit **163** calculates the habit expected value on the basis of the action history data stored in the storage unit **106**. The habit expected value indicates a habitual expected value of the user for a predetermined action of the autonomous mobile body **11**. The habit expected value calculation unit **163** supplies data indicating the calculated habit expected value to the total expected value calculation unit **164**.

[0112] The total expected value calculation unit **164** calculates a total expected value that is a final expected value, on the basis of the promotion expected value, the suppression expected value, and the habit expected value. The total expected value calculation unit **164** supplies the calculated total expected value to the action planning unit **123**.

[0113] The action planning unit **123** plans the action of the autonomous mobile body **11** on the basis of, for example, the total expected value.

<Functional Configuration Example of Information Processing Terminal **12**>

[0114] Next, a functional configuration example of the information processing terminal **12** will be described with reference to FIG. **8**. The information processing terminal **12** includes an input unit **201**, a communication unit **202**, an information processing unit **203**, an output unit **204**, and a storage unit **205**.
[0115] The input unit **201** includes, for example, input devices such as a switch and a button. Furthermore, the input unit **201** includes various sensors, for example, a camera, a microphone, an inertial sensor, and the like. The input unit **201** supplies input data input via the input device and sensor data output from the various sensors to the information processing unit **203**.
[0116] The communication unit **202** communicates with the autonomous mobile body **11**, another information processing terminal **12**, and the information processing server **13** via the network **21** or without the network **21**, and transmits and receives various data. The communication unit **202** supplies the received data to the information processing unit **203**, and acquires the data to be transmitted from the information processing unit **203**.
[0117] Note that, a communication method of the communication unit **202** is not particularly limited, and can be flexibly changed depending on the specifications and operation.
[0118] The information processing unit **203** includes, for example, a processor such as a CPU, performs various types of information processing, and controls each unit of the information processing terminal **12**.
[0119] The output unit **204** includes, for example, a display, a speaker, a haptics device, and the like, and outputs visual information, auditory information, tactile information, and the like on the basis of control by the information processing unit **203**.
[0120] The storage unit **205** includes, for example, a non-volatile memory and a volatile memory, and stores various programs and data.
[0121] Note that, the functional configuration of the information processing terminal **12** can be flexibly changed depending on the specifications and operation.
<Functional Configuration Example of Information Processing Server **13**>
[0122] Next, a functional configuration example of the information processing server **13** will be described with reference to FIG. **9**. The information processing server **13** includes a communication unit **301**, an information processing unit **302**, and a storage unit **303**.
[0123] The communication unit **301** communicates with each autonomous mobile body **11** and each information processing terminal **12** via the network **21**, and transmits and receives various data. The communication unit **301** supplies the received data to the information processing unit **302**, and acquires the data to be transmitted from the information processing unit **302**.
[0124] Note that, a communication method of the communication unit **301** is not particularly limited, and can be flexibly changed depending on the specifications and operation.
[0125] The information processing unit **302** includes, for example, a processor such as a CPU, performs various types of information processing, and controls each unit of the information processing terminal **12**. The information processing unit **302** includes an autonomous mobile body control unit **321** and an application control unit **322**.
[0126] The autonomous mobile body control unit **321** has a configuration similar to the information processing unit **103** of the autonomous mobile body **11**. Specifically, the information processing unit **103** includes a recognition unit **331**, a learning unit **332**, an action planning unit **333**, and an operation control unit **334**.
[0127] Then, the autonomous mobile body control unit **321** has a function similar to the information processing unit **103** of the autonomous mobile body **11**. For example, the autonomous mobile body control unit **321** receives sensor data, input data and the like, and the action history data and the like from the autonomous mobile body **11**, and recognizes situations of the autonomous mobile body **11** and its surroundings. For example, the autonomous mobile body control unit **321** generates control data for controlling the operation of the autonomous mobile body **11** on the basis of the situations of the autonomous mobile body **11** and the surroundings, and

transmits the control data to the autonomous mobile body **11** to control the operation of the autonomous mobile body **11**. For example, the autonomous mobile body control unit **321** performs the pattern recognition learning and the learning of the action pattern corresponding to the user's discipline, similarly to the autonomous mobile body **11**.

[0128] Note that, the learning unit **332** of the autonomous mobile body control unit **321** can also perform learning of collective knowledge common to a plurality of the autonomous mobile bodies **11** by performing the pattern recognition learning and the learning of the action pattern corresponding to the user's discipline on the basis of data collected from the plurality of autonomous mobile bodies **11**.

[0129] The application control unit **322** communicates with the autonomous mobile body **11** and the information processing terminal **12** via the communication unit **301**, and controls the application executed by the information processing terminal **12**.

[0130] For example, the application control unit **322** collects various data regarding the autonomous mobile body **11** from the autonomous mobile body **11** via the communication unit **301**. Then, the application control unit **322** transmits the collected data to the information processing terminal **12** via the communication unit **301**, thereby causing the application executed by the information processing terminal **12** to display the data regarding the autonomous mobile body **11**.

[0131] For example, the application control unit **322** receives data indicating an instruction to the autonomous mobile body **11** input via the application from the information processing terminal **12** via the communication unit **301**. Then, the application control unit **322** gives an instruction from the user to the autonomous mobile body **11** by transmitting the received data to the autonomous mobile body **11** via the communication unit **301**.

[0132] The storage unit **303** includes, for example, a non-volatile memory and a volatile memory, and stores various programs and data.

[0133] Note that, the functional configuration of the information processing server **13** can be flexibly changed depending on the specifications and operation.

<Processing of Information Processing System **1**>

[0134] Next, processing of the information processing system **1** will be described with reference to FIGS. **10**A, **10**B, **11**A, **11**B, **12**A, **12**B, **13**A, **13**B, **14**, **15**A, **15**B, **16**, **17**, **18**, **19**, **20**, **21**, **22**, **23**, **24**, and **25**.

<Pattern Recognition Learning Processing>

[0135] First, pattern recognition learning processing executed by the information processing system **1** will be described with reference to FIGS. **10**A, **10**B, **11**A, **11**B, **12**A, **12**B, **13**A, **13**B, **14**, **15**A, **15**B, and **16**.

[0136] Hereinafter, a case where the pattern recognition learning is performed by the learning unit **122** of the autonomous mobile body **11** will be described as a main example. Note that, the pattern recognition learning may be performed by, for example, the learning unit **323** of the information processing server **13**, or may be performed independently or in cooperation by both the learning unit **122** and the learning unit **323**.

[0137] First, an outline of pattern recognition learning based on teaching will be described with reference to FIGS. **10**A and **10**B. Note that, FIGS. **10**A and **10**B illustrate an example in which the teaching is executed by the user.

[0138] The teaching is performed by a gesture such as pointing by the user, or an utterance, for example, as illustrated in FIG. **10**A. In the case of this example, the user indicates an object O**1** that is a “vacuum cleaner” by a finger UH, and teaches the autonomous mobile body **11** that the object O**1** is the “vacuum cleaner” by an utterance UO.

[0139] At this time, the recognition unit **121** first detects a start of the teaching on the basis of the user's utterance such as “Remember”, the user's gesture indicating a start of learning by the teaching, or a sound such as snapping the user's fingers. Furthermore, at this time, the operation control unit **124** may cause the autonomous mobile body **11** to perform an operation indicating that learning based on the teaching is started. For example, the operation control unit **124** may cause the autonomous mobile body **11** to bark, raise the ears or the tail, or change the color of the irises expressed by the displays **51**. Furthermore, in a case where the autonomous mobile body **11** communicates with the user by using a language, it is also possible to cause the output unit **105** to output a voice to the effect that the learning is disclosed.

[0140] Next, the operation control unit **124** controls the drive unit **104** to change a position and a posture of the autonomous mobile body **11** so that a user's finger UF and the object O**1** indicated by the finger UF are within a field of view FV.

[0141] Subsequently, the recognition unit **121** specifies the object O**1** as a learning target on the basis of a direction indicated by the user's finger UF. Furthermore, the operation control unit **124** causes the input unit **101** to capture an image of the object O**1** specified as the learning target by the recognition unit **121**.

[0142] Furthermore, the recognition unit **121** extracts a noun phrase “vacuum cleaner” used as a label by performing morphological analysis on the user's utterance UO.

[0143] Subsequently, the learning unit **122** associates the label extracted as described above with the captured image of the learning target to obtain learning data, and executes object recognition learning related to the object O**1**.

[0144] As described above, according to the autonomous mobile body **11**, it is possible to automatically collect learning data related to various objects in daily life and perform object recognition learning based on the learning data without preparing learning data in advance.

[0145] Furthermore, according to the above-described function of the autonomous mobile body **11**, by repeatedly executing the learning based on the teaching, or using the collective knowledge learned by the information processing server **13**, it is possible to recognize an object O**2** whose shape (characteristic) is different from that of the object O**1** as “vacuum cleaner”, for example, as illustrated in FIG. **10**B. As a result, for example, even in a case where the user purchases a new vacuum cleaner for replacement, the autonomous mobile body **11** can flexibly respond without performing learning again from scratch.

[0146] Note that, in FIGS. **10**A and **10**B, the object recognition learning is mentioned as an example of the pattern recognition learning, but the pattern recognition learning is not limited to such an example. The pattern recognition learning includes, for example, motion recognition learning. That is, the learning target may be a motion of an object.

[0147] FIGS. **11**A and **11**B are diagrams illustrating an outline of motion recognition learning based on teaching. In the example illustrated in FIGS. **11**A and **11**B, a case is illustrated where the user teaches the autonomous mobile body **11** a motion of “jump” performed by a person.

[0148] Specifically, as illustrated in FIG. **11**A, the user performs an utterance UO for teaching that a motion performed by a person P**1** is “jump” while indicating the person P**1** performing the “jump” with the finger UH.

[0149] At this time, the recognition unit **121** may detect that the user teaches the motion performed by the person P**1** instead of the object recognition (or user recognition) related to the person P**1** by recognizing a phrase “motion” included in the utterance UO. Furthermore, the recognition unit **121** may detect the teaching related to the motion recognition on the basis of an utterance such as “Remember the motion” uttered by the user.

[0150] Subsequently, the recognition unit **121** specifies the motion performed by the person P**1** as a learning target on the basis of a direction indicated by the user's finger UF. Furthermore, the operation control unit **124** causes the input unit **101** to capture an image of the motion of the person P**1** specified as the learning target by the recognition unit **121**.

[0151] Furthermore, the recognition unit **121** extracts a noun phrase “jump” used as a label by performing morphological analysis on the user's utterance UO.

[0152] Subsequently, the learning unit **122** associates the label extracted as described above with the captured image of the learning target to obtain learning data, and executes motion recognition learning related to the motion performed by the person P**1**.

[0153] As described above, according to the autonomous mobile body **11**, it is possible to automatically collect learning data related to various motions performed by the object in addition to the object itself, and perform motion recognition learning based on the learning data.

[0154] Note that, in the above, a case has been described where the image of the motion is used as the learning data, as an example; however, the learning unit **122** may use, for example, motion data collected by the information processing terminal **12** worn by the person performing the motion as the learning data.

[0155] According to the above-described function of the autonomous mobile body **11**, by repeatedly executing the learning based on the teaching, or using the collective knowledge learned by the information processing server **13**, it is also possible to recognize "jump" or the like performed by a person P**2** different from the person P**1** with high accuracy, as illustrated in FIG. **11**B.

[0156] Furthermore, the pattern recognition learning may include, for example, spatial area recognition learning. That is, the learning target may be any spatial area. Here, the spatial area may be any predetermined area (place) in the space. Note that, the spatial area does not necessarily have to be a closed space physically separated from other spatial areas. The spatial area may be, for example, a "house" or the "first floor" of the "house". Furthermore, the spatial area may be a "living room" on the "first floor" or a "near a sofa" in the "living room".

[0157] FIGS. **12**A and **12**B are diagrams illustrating an outline of spatial area recognition learning based on teaching. In the example illustrated in FIGS. **12**A and **12**B, a case is illustrated where the user teaches the autonomous mobile body **11** a spatial area D**1** that is an "entrance".

[0158] Specifically, as illustrated in FIG. **12**A, the user performs an utterance UO that teaches that the spatial area D**1** in which the autonomous mobile body **11** is located is the "entrance".

[0159] At this time, the recognition unit **121** may detect that teaching related to the spatial area D**1** is being performed, by recognizing a phrase "here" included in the utterance UO. Furthermore, the recognition unit **121** may detect the teaching related to the spatial area recognition on the basis of an utterance such as "Remember the place" uttered by the user.

[0160] Subsequently, the recognition unit **121** specifies the spatial area D**1** in which the autonomous mobile body **11** is currently located as a learning target on the basis of the user's utterance UO. Furthermore, the operation control unit **124** causes the input unit **101** to capture an image of the spatial area specified as the learning target by the recognition unit **121**.

[0161] Furthermore, the recognition unit **121** extracts a noun phrase "entrance" used as a label by performing morphological analysis on the user's utterance UO.

[0162] Subsequently, the learning unit **122** associates the label extracted as described above with the captured image of the learning target to obtain learning data, and executes spatial area recognition learning related to the "entrance".

[0163] As described above, according to the autonomous mobile body **11**, it is possible to automatically collect learning data related to various spatial areas in addition to objects and motions, and perform spatial area recognition learning based on the learning data.

[0164] Note that, in the above, a case has been described where the image of the spatial area is used as the learning data, as an example; however, the learning unit **122** may use, as a feature of the spatial area to be learned, for example, a fact that the user who is being tracked in the spatial area D**1** often disappears (that is, goes out), or a fact that utterances such as "I'm home" and "See you" are often detected in the spatial area D**1**.

[0165] According to the above-described function of the autonomous mobile body **11**, by repeatedly executing the learning based on the teaching, or using the collective knowledge learned by the information processing server **13**, it is possible to recognize a spatial area D**2** with a different taste from that of the spatial area D**1**, as the "entrance", as illustrated in FIG. **12**B.

[0166] Here, as described above, the operation control unit **124** has a function of causing the input unit **101** to capture an image of the learning target specified by the recognition unit **121**.

[0167] At this time, the operation control unit **124** may control the input unit **101** and the drive unit **104** so that the pattern recognition is performed with high accuracy and efficiency. For example, the operation control unit **124** can change the position and posture of the autonomous mobile body **11** so that an image of the entire specified object is correctly captured by controlling the drive unit **104**.

[0168] As a result, efficient pattern recognition learning can be performed on the basis of an image captured at an appropriate distance from the learning target.

[0169] Furthermore, the operation control unit **124** may control the drive unit **104** and the input unit **101** so that an image of the learning target specified by the recognition unit **121** is captured from a plurality of angles.

[0170] As a result, features of the learning target can be extracted from various angles as compared with a case where learning is performed based on an image of one aspect of the learning target, and a learning effect with high generalization can be obtained.

[0171] Furthermore, for example, the teaching may be implemented by a marker such as a QR code (registered trademark) attached to the learning target. FIGS. **13**A and B are diagrams for explaining teaching using a marker.

[0172] For example, FIG. **13**A illustrates an example in which teaching related to the object recognition learning is implemented with a marker M**1** attached to the object O**1** that is the “vacuum cleaner”. In this case, the recognition unit **121** can acquire a label “vacuum cleaner” on the basis of an image of the marker M**1** captured by the input unit **101**.

[0173] Furthermore, FIG. **13**B illustrates an example in which teaching related to the spatial area recognition learning is implemented with a marker M**2** attached to a door installed in the spatial area D**1** that is the “entrance”. In this case as well, the recognition unit **121** can acquire a label “entrance” on the basis of an image of the marker M**2** captured by the input unit **101**.

[0174] As described above, by using the marker, it is possible to implement the teaching related to the object, the spatial area, and the like, instead of explicit teaching by the user, and it is possible to automatically enhance recognition ability of the autonomous mobile body **11**.

[0175] Moreover, for example, the teaching may be performed on the basis of information transmitted from the learning target by wireless communication. FIG. **14** is a diagram for explaining teaching using wireless communication.

[0176] In the case of the example illustrated in FIG. **14**, the object O**1** that is the “vacuum cleaner” transmits the label “vacuum cleaner” and an image I**3***a* and an image I**3***b* of the object O**1** to the autonomous mobile body **11** by wireless communication. At this time, the learning unit **122** can perform object recognition learning related to the object O**1** on the basis of the received label “vacuum cleaner”, the image I**3***a*, and the image I**3***b*.

[0177] According to the teaching using wireless communication as described above, for example, as illustrated in the figure, even if the object O**3** that is a learning target is stored in a closet and an image of the object O**3** cannot be captured, by transmitting an image prepared in advance to the autonomous mobile body **11** together with the label, it is possible for the autonomous mobile body **11** to perform object recognition learning related to the object O**3**.

[0178] For wireless communication, for example, Near Field Communication (NFC), Bluetooth (registered trademark), Radio Frequency Identification (RFID), beacon, and the like are used.

[0179] Furthermore, for example, the teaching may be implemented by an inaudible sound such as an ultrasonic wave emitted by a learning target. FIGS. **15**A and **15**B are diagrams for explaining teaching using the inaudible sound.

[0180] In the example illustrated in FIGS. **15**A and **15**B, an example is illustrated in which an object O**5** that is a “washing machine” emits a predetermined inaudible sound during operation, whereby the teaching is implemented. For example, FIG. **15**A illustrates an example of a case where the recognition unit **121** detects that the object O**5** is the “washing machine” and the object

O**5** is “during dewatering”, on the basis of an inaudible sound NAS**1** emitted by the object O**5** “during dewatering”.
[0181] For example, FIG. **15**B illustrates an example of a case where the recognition unit **121** detects that the object O**5** is the “washing machine” and the object O**5** is “during drying”, on the basis of an inaudible sound NAS**2** emitted by the object O**5** “during drying”.
[0182] As described above, according to the teaching using the inaudible sound, it is possible to teach not only a name of an object but also an operating state and the like. Furthermore, according to the teaching using the inaudible sound, for example, the learning unit **122** can also learn an audible sound AS**1** that is an operation sound emitted by the object O**5** “during dewatering” and an audible sound AS**2** that is an operation sound emitted by the object O**5** “during drying”, together with an operating state of the object O**5**. By repeatedly executing the learning as described above, the autonomous mobile body **11** can gradually acquire the recognition ability even for an object that does not emit an inaudible sound.
[0183] Next, correction of association between the label and the learning target will be described. As described above, the autonomous mobile body **11** can perform pattern recognition learning on the basis of various types of teaching.
[0184] However, for example, when learning is performed on the basis of teaching by the user, a situation is assumed where the learning target or the label is erroneously acquired. For this reason, the application control unit **322** of the information processing server **13** may provide the information processing terminal **12** with a user interface for the user (or a developer or a service provider) to correct the association between the label and the learning target.
[0185] Furthermore, the correction of a learning result may be automatically performed by the autonomous mobile body **11**. For example, by comparing the learning result accumulated in the information processing server **13** with the learning result of the autonomous mobile body **11**, the learning unit **122** can also detect a discrepancy between the collective knowledge and learning knowledge of the autonomous mobile body **11**, and automatically correct the label and the learning target.
[0186] Moreover, for example, the operation control unit **124** may cause the autonomous mobile body **11** to perform a guiding operation for guiding the teaching by the user to collect learning data more effectively.
[0187] For example, in a case where the recognition unit **121** detects an unknown object that cannot be recognized during an autonomous action of the autonomous mobile body **11**, the operation control unit **124** can cause the autonomous mobile body **11** to perform a guiding operation for guiding the user's teaching for the unknown object. For example, the operation control unit **124** causes the autonomous mobile body **11** to perform a barking operation to the unknown object as the guiding operation.
[0188] As described above, with the guiding operation, it is possible to increase a possibility that the user performs the teaching for the unknown object with respect to an action of the autonomous mobile body **11**, and an effect is expected of collecting learning data more efficiently.
[0189] Note that, examples of the guiding operation include various operations such as barking, smelling, intimidating, tilting the head, alternately looking at the user and a target, and being frightened.
[0190] Next, a flow of the pattern recognition learning processing of the autonomous mobile body **11** described above will be described with reference to a flowchart of FIG. **16**.
[0191] In step S**1**, the recognition unit **121** detects a start of teaching. The recognition unit **121** detects the start of teaching on the basis of, for example, an utterance by the user, detection of a QR code (registered trademark), reception of information by wireless communication, detection of an inaudible sound, or the like.
[0192] In step S**2**, the operation control unit **124** performs notification of a start of learning. For example, the operation control unit **124** causes the autonomous mobile body **11** to perform an

operation indicating that pattern recognition learning is started. For example, the operation control unit **124** causes the autonomous mobile body **11** to perform operations such as barking, moving the ears and the tail, and changing the color of the iris.

[0193] In step S**3**, the recognition unit **121** specifies a learning target. The recognition unit **121** may specify the learning target on the basis of a gesture such as pointing by the user, for example, or may specify the learning target on the basis of information acquired from the learning target.

[0194] In step S**4**, the recognition unit **121** acquires a label. For example, the recognition unit **121** may extract the label from the user's utterance, or may acquire the label from the information acquired from the learning target.

[0195] In step S**5**, the operation control unit **124** controls the drive unit **104** so that a position and a posture are obtained in which an image of the entire learning target specified in step S**3** can be captured.

[0196] In step S**6**, the operation control unit **124** controls the drive unit **104** and the input unit **101** so that an image of the learning target is captured from a plurality of angles.

[0197] In step S**7**, the learning unit **122** executes the pattern recognition learning based on the label acquired in the processing of step S**4** and the image captured in the processing of step S**6**.

[0198] In step S**8**, the operation control unit **124** performs notification of completion of learning. For example, the operation control unit **124** causes the autonomous mobile body **11** to perform an operation indicating the completion of the pattern recognition learning. The operation may be, for example, barking, moving the ears or the tail, changing the color of the iris, or the like.

[0199] Thereafter, the pattern recognition learning processing ends.

[0200] As described above, the autonomous mobile body **11** can improve the recognition ability to recognize an object, a motion, a spatial area, and the like by performing the pattern recognition learning.

<Discipline Response Processing for Autonomous Mobile Body

[0201] Next, discipline response processing executed by the information processing system **1** will be described with reference to FIGS. **17** to **25**.

[0202] The discipline response processing is processing in which the autonomous mobile body **11** changes its action in response to discipline by the user.

[0203] Here, as a target to be recognized or stored by the autonomous mobile body **11** in the discipline response processing, for example, a situation in which the autonomous mobile body **11** is placed described above is assumed. Note that, a person to be recognized or stored by the autonomous mobile body **11** is assumed to be, for example, the user of the autonomous mobile body **11**, a user of another autonomous mobile body **11**, a stranger, or the like. The user of the autonomous mobile body **11** includes, for example, an owner of the autonomous mobile body **11**, the owner's family, relatives, and the like. Furthermore, autonomous mobile bodies such as targets to be recognized or stored by the autonomous mobile body **11** include not only an autonomous mobile body of the same type as the autonomous mobile body **11** but also autonomous mobile bodies of other types.

[0204] As how to discipline the autonomous mobile body **11**, for example, a method is assumed in which the user gives an instruction by voice or gesture, gives an instruction by using an application on the information processing terminal **12**, touches the autonomous mobile body **11**, or gives food. Furthermore, for example, a method is assumed in which the autonomous mobile body **11** is caused to recognize the above-described target, the user's instruction, and the like from various sensors included in the input unit **101** of the autonomous mobile body **11** with use of an image, voice, or the like.

[0205] As a process in which the autonomous mobile body **11** changes by discipline, for example, a process is assumed in which a change of the autonomous mobile body **11** is completed by one discipline, the autonomous mobile body **11** gradually changes by repetition of similar discipline, or the autonomous mobile body **11** changes stepwise by discipline performed in accordance with a

predetermined procedure.
[0206] As a mode of the change of the autonomous mobile body **11**, it is assumed that, for example, behavior changes, a predetermined action is acquired, the predetermined action is improved (proficiency is increased in the predetermined action), or the predetermined action is stopped.
[0207] As a method of expressing the change of the autonomous mobile body **11**, in addition to a method in which the autonomous mobile body changes the action, for example, a method is assumed of displaying information indicating the change of the autonomous mobile body **11** in the application on the information processing terminal **12**.
[0208] Note that, hereinafter, a case where the discipline response processing is performed by the learning unit **122** of the autonomous mobile body **11** will be described as a main example. Note that, the discipline response processing may be performed by, for example, the learning unit **323** of the information processing server **13**, or may be performed independently or in cooperation by both the learning unit **122** and the learning unit **323**.
[0209] Furthermore, although detailed description is omitted, the above-described pattern recognition learning is appropriately performed as necessary in the discipline response processing.
<First Method of Disciplining Autonomous Mobile Body **11**>
[0210] First, a first method of disciplining the autonomous mobile body **11** will be described with reference to FIGS. **17** to **20**.
[0211] In the first method, the user gives an instruction to promote or suppress a predetermined action, whereby the autonomous mobile body **11** is disciplined.
[0212] Note that, a method in which the user gives an instruction is not particularly limited as long as the autonomous mobile body **11** can recognize the user's instruction. Hereinafter, a description will be given mainly with an example in which the user gives an instruction to the autonomous mobile body **11** by voice.
[0213] Furthermore, hereinafter, as a specific example, a case will be described where the autonomous mobile body **11** is disciplined to behave quietly when the user wants the autonomous mobile body **11** to be quiet.
<First Embodiment of Action Control Processing>
[0214] Here, a first embodiment of action control processing executed by the autonomous mobile body **11** will be described with reference to a flowchart of FIG. **17**.
[0215] The processing is started, for example, when the power of the autonomous mobile body **11** is turned on, and ends when the power of the autonomous mobile body **11** is turned off.
[0216] In step S**101**, the recognition unit **121** determines whether or not an instruction is given from the user. Specifically, the recognition unit **121** performs processing of recognizing the user's instruction on the basis of the sensor data and input data supplied from the input unit **101** and the received data supplied from the communication unit **102**. In a case where the recognition unit **121** does not recognize the user's instruction, the recognition unit **121** determines that no instruction is given from the user, and the processing proceeds to step S**102**.
[0217] In step S**102**, the recognition unit **121** determines whether or not it is a situation to start an action. For example, the recognition unit **121** recognizes a current situation on the basis of the sensor data and input data supplied from the input unit **101** and the received data supplied from the communication unit **102**.
[0218] Furthermore, the recognition unit **121** reads, from the storage unit **106**, action situation correlation data obtained by learning a correlation between each action and each situation in the processing of step S**110** described later.
[0219] The action situation correlation data is one of the learning result data, and is data indicating a correlation between each action and a situation in which each action should be executed (hereinafter referred to as an action trigger situation). For example, the action situation correlation data indicates a correlation between a situation in which an instruction is given by the user and an action for which the instruction is given.

[0220] A detail of the action trigger situation can be set arbitrarily. For example, the action trigger situation includes several items of the time, time zone, day of the week, place, state of the user, state of the autonomous mobile body **11**, situation around the autonomous mobile body **11**, and the like in which the action should be taken.
[0221] Furthermore, types of the items included in the action trigger situation may differ depending on a detail of a corresponding action. For example, the action trigger situation in a case where the detail of the action is “be quiet” is represented by the situation around the autonomous mobile body **11**, and the action trigger situation in a case where the detail of the action is “sleep” is represented by the time.
[0222] In a case where there is no action trigger situation similar to the current situation in the action situation correlation data, the recognition unit **121** determines that it is not the situation to start the action, and the processing returns to step S**101**.
[0223] Thereafter, the processing of steps S**101** and S**102** is repeatedly executed until it is determined in step S**101** that the instruction is given from the user, or it is determined in step S**102** that it is the situation to start the action.
[0224] On the other hand, in step S**101**, in a case where the recognition unit **121** recognizes the user's instruction, the recognition unit **121** determines that the instruction is given from the user, and the processing proceeds to step S**103**.
[0225] For example, in a case where the recognition unit **121** recognizes the user's voice prompting the autonomous mobile body **11** to be quiet, such as “Be quiet”, “Shut up”, or “Noisy”, the recognition unit **121** determines that the instruction is given from the user, and the process proceeds to step S**103**.
[0226] In step S**103**, the recognition unit **121** recognizes the current situation on the basis of the sensor data and input data supplied from the input unit **101** and the received data supplied from the communication unit **102**. That is, the recognition unit **121** recognizes a situation when the instruction is given from the user. Details of the situation to be recognized include, for example, some of the time, time zone, day of the week, place, state of the user, state of the autonomous mobile body **11**, situation around the autonomous mobile body **11**, and the like when the instruction is given.
[0227] The recognition unit **121** supplies data indicating a detail of the user's instruction to the action planning unit **123**. Furthermore, the recognition unit **121** supplies, to the learning unit **122**, data indicating the detail of the user's instruction and the situation when the instruction is given.
[0228] In step S**104**, the action planning unit **123** sets the action detail on the basis of the number of times of action taken in the past in accordance with the user's instruction.
[0229] Specifically, the action planning unit **123** specifies the number of times of action (hereinafter, referred to as the number of times of instruction action) taken in the past in accordance with the instruction of a similar detail on the basis of the action history data stored in the storage unit **106**. Note that, the number of times of instruction action is, for example, the number of times the action for which the instruction is given is executed to the end and completed, and does not include the number of times the action for which the instruction is given is stopped in the middle and not completed.
[0230] The action planning unit **123** sets the action detail on the basis of the specified number of times of instruction action. For example, the action planning unit **123** sets the action detail so that as the number of times of instruction action increases, a state of being proficient in the action for which the instruction is given can be expressed.
[0231] Specifically, in a case where the number of times of instruction action is 0 times, for example, in a case where the autonomous mobile body **11** is not instructed to be quiet in the past and is instructed for the first time this time, the action detail is set to a pattern of sitting in a restless state for 1 minute.
[0232] For example, in a case where the number of times of instruction action is 1 to 10 times, the

action detail is set to a pattern of sitting for the number of times of instruction action×1 minutes and then sitting in a restless state for 1 minute.
[0233] For example, in a case where the number of times of instruction action is 11 to 20 times, the action detail is set to a pattern of sitting for 10 minutes, lying down for (the number of times of instruction action−10)×1 minutes, and then sitting in a restless state for 1 minute.
[0234] For example, in a case where the number of times of instruction action is 21 to 30 times, the action detail is set to a pattern of sitting for 10 minutes, lying down for 10 minutes, sleeping for (the number of times of instruction action−20)×1 minutes, and then sitting in a restless state for 1 minute.
[0235] For example, in a case where the number of times of instruction action is 31 times or more, the action detail is set to a pattern of sitting for 10 minutes, lying down for 10 minutes, sleeping for 10 minutes, and then sitting in a restless state for 1 minute.
[0236] As the instruction for the predetermined action is repeated in this way, the detail of the predetermined action changes. In the case of this example, as the number of times of instruction action increases, a proficiency level for the action to be quiet (hereinafter referred to as a silent action) of the autonomous mobile body **11** improves. That is, the time during which the autonomous mobile body **11** can stand by quietly becomes longer. As a result, the user can feel the growth of the autonomous mobile body **11**.
[0237] The action planning unit **123** supplies data indicating the set action detail to the operation control unit **124**.
[0238] Thereafter, the processing proceeds to step S**106**.
[0239] On the other hand, in step S**102**, in a case where there is an action trigger situation similar to the current situation in the action situation correlation data, the recognition unit **121** determines that it is the situation to start the action, and the processing proceeds to step S**105**.
[0240] For example, in a case where an instruction to be quiet is repeatedly given when the user is at a desk to the autonomous mobile body **11** in the past, a situation in which the user is at the desk is associated with the silent action, as an action trigger situation. In this case, in a case where the user is currently at the desk, it is determined that it is the situation to start the action, and the processing proceeds to step S**105**.
[0241] For example, in a case where an instruction to be quiet is repeatedly given at the same time zone to the autonomous mobile body **11** in the past, the time zone is associated with the silent action, as an action trigger situation. In this case, in a case where a current time is included in that time zone, it is determined that it is the situation to start the action, and the processing proceeds to step S**105**.
[0242] In step S**105**, the autonomous mobile body **11** sets the action detail on the basis of the current situation.
[0243] Specifically, the recognition unit **121** notifies the action planning unit **123** of the action trigger situation determined to be similar to the current situation. Furthermore, the recognition unit **121** supplies data indicating the recognized current situation to the learning unit **122**.
[0244] The action planning unit **123** sets the action detail on the basis of the action trigger situation of which the notification is performed and the action situation correlation data stored in the storage unit **106**. That is, an action detail associated with the action trigger situation of which the notification is performed in the action situation correlation data is set as the action detail to be executed by the autonomous mobile body **11**. The action planning unit **123** supplies data indicating the set action detail to the operation control unit **124**.
[0245] Thereafter, the processing proceeds to step S**106**.
[0246] In step S**106**, the autonomous mobile body **11** starts the action. Specifically, the operation control unit **124** starts processing of controlling the drive unit **104** and the output unit **105** to execute the set action detail.
[0247] In step S**107**, the operation control unit **124** determines whether or not the action is

completed. In a case where the action set in the processing of step S**104** or step S**105** is still being executed, the operation control unit **124** determines that the action is not completed yet, and the processing proceeds to step S**108**.
[0248] In step S**108**, the action planning unit **123** determines whether or not to stop the action. For example, the recognition unit **121** performs processing of recognizing the user's instruction on the basis of the sensor data and input data supplied from the input unit **101** and the received data supplied from the communication unit **102**. Then, in a case where the recognition unit **121** does not recognize an instruction that triggers to stop the action currently being executed, the action planning unit **123** determines not to stop the action, and the processing returns to step S**107**.
[0249] Thereafter, the processing of steps S**107** and S**108** is repeatedly executed until it is determined in step S**107** that the action is completed, or it is determined in step S**108** to stop the action.
[0250] On the other hand, in a case where it is determined in step S**107** that the action is completed, the processing proceeds to step S**110**.
[0251] Furthermore, in step S**108**, for example, in a case where it is recognized that the instruction that triggers to stop the action currently being executed, the recognition unit **121** notifies the action planning unit **123** that the instruction that triggers to stop the action currently being executed is recognized. Then, the action planning unit **123** determines to stop the action, and the processing proceeds to step S**110**.
[0252] As the instruction that triggers to stop the action, for example, an instruction for suppressing the action being executed, an instruction for another action different from the action being executed, and the like are assumed.
[0253] For example, in a case where the autonomous mobile body **11** is performing a silent action, in a case where the user calls the name of the autonomous mobile body **11** or gives an instruction such as “Come here” or “You may move”, it is determined to stop the action, and the processing proceeds to step S**110**.
[0254] In step S**110**, the autonomous mobile body **11** stops the action. Specifically, the recognition unit **121** supplies data indicating a detail of the instruction that triggers to stop the action to the action planning unit **123**. The action planning unit **123** instructs the operation control unit **124** to stop the action. The operation control unit **124** controls the drive unit **104** and the output unit **105** to cause the autonomous mobile body **11** to stop the action currently being executed.
[0255] Thereafter, the processing proceeds to step S**110**.
[0256] In step S**110**, the learning unit **122** learns a correlation between the action and the situation.
[0257] Specifically, the action planning unit **123** supplies the executed action detail and data indicating whether or not the action is completed to the learning unit **122**.
[0258] The learning unit **122** adds a result of the action of this time to the action history data stored in the storage unit **106**.
[0259] Furthermore, the learning unit **122** updates the action situation correlation data by learning the correlation between each action and the situation on the basis of the action history data. For example, the learning unit **122** calculates a degree of correlation (for example, a correlation coefficient) between each action and each situation.
[0260] Note that, for example, in a case where a certain action is completed, the degree of correlation between the action and a situation when the action is performed is made large. On the other hand, for example, in a case where a certain action is stopped in the middle, the degree of correlation between the action and a situation when the action is performed is made small.
[0261] Then, the learning unit **122** associates a situation in which the degree of correlation is greater than or equal to a predetermined threshold value with each action, as an action trigger situation.
[0262] Thus, for example, in a case where the user repeatedly gives a similar instruction in a similar situation in the past, the situation is associated with an action corresponding to the

instruction, as an action trigger situation.
[0263] For example, in a situation in which the user is at the desk, in a case where the user repeatedly gives an instruction “Be quiet” to the autonomous mobile body **11**, the situation in which the user is at the desk is associated with the silent action, as a trigger situation.
[0264] For example, in a case where the user repeatedly gives an instruction “Be quiet” to the autonomous mobile body **11** in a similar time zone, the time zone is associated with the silent action, as a trigger situation.
[0265] On the other hand, the action trigger situation is not associated with an action for which there is no situation in which the degree of correlation is greater than or equal to the threshold value.
[0266] The learning unit **122** stores the updated action situation correlation data in the storage unit **106**.
[0267] Thereafter, the processing returns to step S**101**, and the processing after step S**101** is executed.
[0268] As described above, the user can experience the discipline for the autonomous mobile body **11**. As a result, for example, the user can feel that the autonomous mobile body **11** grows due to the user's discipline, and the user experience is improved.
[0269] Furthermore, the autonomous mobile body **11** comes to be able to voluntarily execute an appropriate action under an appropriate situation without the user's instruction by learning the situation when the instruction is given from the user.
[0270] Note that, for example, the autonomous mobile body **11** can estimate a time series transition of the user's expected value for the action for which the instruction is given on the basis of a distribution of time at which the instruction is given from the user, or the like, and can change the action on the basis of the estimated expected value.
[0271] Here, with reference to FIGS. **18** to **20**, a specific example will be described of processing of changing the action of the autonomous mobile body **11** on the basis of the user's expected value.
[0272] Note that, hereinafter, an example will be described of a case where the autonomous mobile body **11** performs a silent action in accordance with the user's instruction. Furthermore, expected value calculation processing described below is executed by the expected value calculation unit **151** of FIG. **7**.
[0273] FIG. **18** illustrates an example of time series changes of the silent action, promotion expected value, suppression expected value, and habit expected value of the autonomous mobile body **11**.
[0274] At time t**1**, the user gives an instruction for the silent action. For example, the user calls out “Be quiet” to the autonomous mobile body **11**. As a result, the autonomous mobile body **11** starts the silent action. Furthermore, the promotion expected value rises from 0 to 1.0 and is held until time t**3**.
[0275] At time t**2**, the autonomous mobile body **11** ends the silent action. Note that, as described above, the length of the silent action period between the time t**1** and the time t**2** changes depending on the number of times of instruction action for the past silent action.
[0276] At the time t**3**, the promotion expected value begins to decrease. The promotion expected value decreases linearly.
[0277] Note that, the length between the time t**1** and the time t**3**, that is, the length of a period during which the promotion expected value is held at 1.0 may be constant, or may change depending on the number of times of instruction action for the past silent action. In the latter case, the period becomes longer as the number of times of instruction action for the silent actions increases.
[0278] At time t**4**, the promotion expected value reaches 0.5 and then is lowered to 0.
[0279] At time t**5**, the user gives an instruction for the silent action as at the time t**1**. As a result, the autonomous mobile body **11** starts the silent action, the promotion expected value rises from 0 to

1.0, and is held until time t**7**.
[0280] At time t**6**, the autonomous mobile body **11** ends the silent action as at the time t**2**.
[0281] At the time t**7**, the promotion expected value begins to decrease as at the time t**3**.
[0282] At time t**8**, the user gives an instruction to suppress silent action. For example, the user calls the name of the autonomous mobile body **11**. As a result, the suppression expected value rises from 0 to 1.0 and then begins to decrease. The suppression expected value decreases linearly.
[0283] At time t**9**, the promotion expected value reaches 0.5 and then is lowered to 0 as at the time t**4**.
[0284] At time t**10**, the suppression expected value reaches 0.5 and then is lowered to 0.
[0285] Note that, the habit expected value changes on the basis of a result calculated by a method described later.
[0286] FIG. **19** is a graph for explaining a method of calculating the total expected value in a case where the promotion expected value, the suppression expected value, and the habit expected value change as illustrated in FIG. **18**.
[0287] At the time t**1**, the instruction for the silent action given from the user serves as a promotion trigger, and a total expected value 1 is valid. Note that, the total expected value 1 is set to a larger value of the promotion expected value and the habit expected value.
[0288] At the time t**5**, the instruction for the silent action given from the user serves as the promotion trigger. However, since the total expected value 1 is already made valid at the time t**1**, the total expected value 1 remains to be made valid.
[0289] At the time t**7**, the instruction to suppress the silent action given from the user serves as a suppression trigger, the total expected value 1 is invalid, and a total expected value 2 is valid. The total expected value 2 is set to a smaller value of (1.0—the suppression expected value) and the habit expected value.
[0290] Thus, in a period from the time t**1** to the time t**7**, the autonomous mobile body **11** acts on the basis of the total expected value 1. For example, in a case where the total expected value 1 is greater than or equal to a predetermined threshold value, the autonomous mobile body **11** performs an action similar to the silent action and operates as quiet as possible, except for a period during which the silent action is executed (a period from the time t**1** to the time t**2**, and a period from the time t**5** to the time t**6**). For example, the autonomous mobile body **11** limits a walking speed, and walks quietly. Furthermore, for example, the autonomous mobile body **11** preferentially performs a quiet operation and tries not to perform a noisy operation as much as possible. The quiet operation is, for example, an operation of operating only the front legs and neck without moving. Noisy operation is, for example, running around, or singing. On the other hand, the autonomous mobile body **11** performs a normal operation in a case where the total expected value 1 is less than the predetermined threshold value.
[0291] As a result, the autonomous mobile body **11** can operate quietly for a while even after finishing the silent action, for example, and the user can feel the afterglow of the silent action.
[0292] Furthermore, in a period after the time t**7**, the autonomous mobile body **11** acts on the basis of the total expected value 2. For example, in a case where the total expected value 2 is greater than or equal to a predetermined threshold value, the autonomous mobile body **11** preferentially performs a quiet operation. On the other hand, the autonomous mobile body **11** performs a normal operation in a case where the total expected value 2 is less than the predetermined threshold value.
[0293] Note that, regarding the promotion trigger and suppression trigger, the latest ones are always valid. Then, as described above, the total expected value 1 is valid during the period when the promotion trigger is valid, and the total expected value 2 is valid during the period when the suppression trigger is valid.
[0294] Here, an example of a method of calculating the habit expected value will be described with reference to FIG. **20**.
[0295] The top graph of FIG. **20** illustrates a histogram of the number of times of activation of the

silent action for each predetermined time zone within an immediately preceding predetermined period (for example, three months). In a case where the silent action is activated even once a day in each time zone, one histogram value is added. That is, in a case where the silent action is activated even once in the same time zone on the same day, one histogram value is added regardless of the number of times of activation. Thus, for example, in a case where the histogram value is five, it indicates that the silent action is activated in the time zone, in five days within the predetermined period.

[0296] Note that, activation of a certain action means that the autonomous mobile body **11** starts the action regardless of the presence or absence of the user's instruction, and it does not matter whether or not the action is completed.

[0297] The second graph of FIG. **20** illustrates a histogram of the number of times of activation of all actions (including the silent action) for each predetermined time zone within the same period as the top graph. For example, in a case where any action is activated even once a day in each time zone, one histogram value is added. That is, in a case where any action is activated even once in the same time zone on the same day, one histogram value is added regardless of the number of times and type of the activated action. Thus, for example, in a case where the histogram value is 65, it indicates that some action is activated in the time zone, in 65 days within the predetermined period.

[0298] Then, a silent action activation rate v(t) in each time zone t is calculated by the following equation (1).

[00001] $v(t) = s(t) / a(t) \quad (1)$

[0299] The number of times of silent action activation in the time zone t is indicated by s(t), and the number of times of all actions activation in the time zone t is indicated by a(t).

[0300] Then, a normalized silent action activation rate vn(t) obtained by normalizing the silent action activation rate v(t) is calculated by the following equation (2).

[00002] $vn(t) = K \times (((v(t) - \quad) / \quad) / 2 + 0.5) \quad (2)$

[0301] An average value of the number of times of silent action activation s(t) in all time zones is indicated by μ, and a variance of the number of times of silent action activation s(t) in all time zones is indicated by σ.

[0302] K indicates a coefficient, and is set as follows, for example. Note that, hereinafter, a total of the number of times of silent action activation s(t) is defined as Σs(t), and a total of the number of times of all actions activation a(t) is defined as Σa(t).

[0303] For example, in a case where Σa(t) is less than a predetermined threshold value MIN_COUNT, that is, in a case where the number of times of activation of the action of the autonomous mobile body **11** is very small, the coefficient K is set to 0.

[0304] On the other hand, in a case where Σa(t) is greater than or equal to the threshold value MIN_COUNT, K is set to 1 when a ratio (=Σs(t)/Σa(t)) is greater than or equal to a predetermined threshold value MAX_RATIO. On the other hand, in a case where the ratio is less than the threshold value MAX_RATIO, K is set to ratio/MAX_RATIO. That is, the coefficient K increases as a rate of the number of times of activation of the silent action to the number of times of activation of all actions increases.

[0305] The graph at the bottom of FIG. **20** illustrates a graph of the normalized silent action activation rate.

[0306] Then, the normalized silent action activation rate after being corrected to 0 in a case where the normalized silent action activation rate is less than 0, or the normalized silent action activation rate after being corrected to 1 in a case where the normalized silent action activation rate is greater than or equal to 1 is used as the habit expected value.

[0307] By using the habit expected value, the autonomous mobile body **11** can perform an action expected by the user (for example, acts quietly) when the user's expectation is high on a daily basis, even if the user does not give an instruction.

[0308] Note that, by the above-described processing, it is possible to discipline the autonomous mobile body **11** to cause various actions other than the silent action to be performed.

[0309] For example, the user speaks (gives an instruction) “Good night” repeatedly (for example, every day), whereby the autonomous mobile body **11** comes to be proficient in a sleeping action. Here, the sleeping action is, for example, an action in which the autonomous mobile body **11** moves to a charging stand and transitions to a standby mode on the charging stand.

[0310] For example, the autonomous mobile body **11** does not easily start the sleeping action even if the user speaks “Good night” at the beginning, but comes to start the sleeping action immediately, after repetition of speaking every day. Furthermore, the autonomous mobile body **11** comes to move quietly to the charging stand, such as walking stealthily.

[0311] Furthermore, the user speaks “Good night” repeatedly in the same time zone, whereby the autonomous mobile body **11** comes to remember the time zone and automatically start the sleeping action in the time zone.

[0312] Moreover, for example, the user speaks “Good night until morning” repeatedly, whereby the autonomous mobile body **11** comes to continue the standby mode until morning in accordance with the user's instruction.

[0313] Furthermore, for example, the user speaks (gives an instruction) “Good morning” repeatedly (for example, every day), whereby the autonomous mobile body **11** comes to be proficient in a wake-up action. Here, the wake-up action is, for example, an action in which the autonomous mobile body **11** is activated from a state in which the autonomous mobile body **11** is in the standby mode on the charging stand and starts moving.

[0314] For example, at the beginning, the autonomous mobile body **11** does not get up easily even if the user speaks “Good morning”, but comes to get up immediately, after repetition of speaking every day.

[0315] Furthermore, the user speaks “Good morning” repeatedly in the same time zone, the autonomous mobile body **11** comes to remember the time zone and automatically wake up in the time zone.

[0316] Moreover, for example, the user speaks (gives an instruction) “I'm home” repeatedly when returning home, whereby the autonomous mobile body **11** comes to be proficient in an action of welcoming the user. Here, the action of welcoming is, for example, an action in which the autonomous mobile body **11** moves to the entrance and welcomes the user.

[0317] For example, at the beginning, the autonomous mobile body **11** does not move to the entrance very often even if the user opens the entrance and speaks “I'm home”, but comes to move to the entrance and welcome the user when the user opens the entrance and speaks “I'm home”, after repetition of speaking every day. Furthermore, for example, the autonomous mobile body **11** comes to remember the user's return time, and automatically move to the entrance and wait for the user to return home when the user's return time comes.

[0318] Furthermore, for example, the user repeatedly gives an instruction to “Wait” in a case where the autonomous mobile body **11** is about to start an operation of eating food, whereby the autonomous mobile body **11** comes to be able to wait without eating food. Furthermore, for example, the user repeatedly gives an instruction to “You may eat” when the autonomous mobile body **11** performs “Wait” for food, whereby the autonomous mobile body **11** comes to be able to perform “Wait” for hood until the user gives permission.

[0319] By a similar method, for example, the autonomous mobile body **11** can be disciplined for “Sitting” and the like.

[0320] Moreover, by a similar method, for example, the user can discipline the autonomous mobile body **11** to suppress a predetermined action by repeatedly giving an instruction to suppress the predetermined action to the autonomous mobile body **11**. For example, in the above-described example, the user can discipline the autonomous mobile body **11** for the silent action by giving a negative instruction such as “Don't be noisy” instead of “Be quiet”.

<Second Method of Disciplining Autonomous Mobile Body **11**>

[0321] Next, a second method of disciplining the autonomous mobile body **11** will be described with reference to FIGS. **21** to **24**.

[0322] In the second method, discipline is performed for a place where the autonomous mobile body **11** performs a predetermined action.

[0323] Hereinafter, as a specific example, an example will be described in which discipline is performed for a place where the autonomous mobile body **11** urinates.

<Second Embodiment of Action Control Processing>

[0324] Here, a second embodiment of the action control processing executed by the autonomous mobile body **11** will be described with reference to a flowchart of FIG. **21**.

[0325] The processing is started, for example, when the power of the autonomous mobile body **11** is turned on, and ends when the power of the autonomous mobile body **11** is turned off.

[0326] In step S**201**, the autonomous mobile body **11** determines whether or not a condition for executing an action is satisfied on the basis of the sensor data and input data supplied from the input unit **101** and the received data supplied from the communication unit **102**. This determination processing is repeatedly executed until it is determined that the condition for executing the action is satisfied, and in a case where it is determined that the condition for executing the action is satisfied, the processing proceeds to step S**202**.

[0327] For example, in a case where the recognition unit **121** recognizes an instruction such as “Pee” by the user, the recognition unit **121** determines that the condition for the action (urination) is satisfied, and the processing proceeds to step S**202**. Furthermore, for example, in a case where the urinary intention coefficient described later becomes greater than or equal to a predetermined threshold value, the action planning unit **123** determines that the condition for the action (urination) is satisfied, and the processing proceeds to step S**202**.

[0328] In step S**202**, the recognition unit **121** determines whether or not a place to act is indicated on the basis of the sensor data and input data supplied from the input unit **101** and the received data supplied from the communication unit **102**. In a case where it is determined that the place to act is indicated, the processing proceeds to step S**203**.

[0329] Note that, the method of indicating the place to act is not particularly limited. For example, the user can indicate the place to urinate by moving the autonomous mobile body **11** to the place to urinate and then uttering a word to indicate the place to urinate, such as “Toilet is here”. For example, the user can indicate the place to urinate by uttering a word to indicate urination and the place to urinate, such as “Pee in the corner of the room”. For example, the user can indicate the place to urinate by indicating a predetermined position on a map by using the information processing terminal **12**.

[0330] Furthermore, for example, the user may indicate a direction of action (for example, “right direction”, “entrance direction”, or the like) instead of a specific place. Moreover, for example, the user may indicate the direction of action and the specific location. Furthermore, for example, the user can indicate the place to act by calling the autonomous mobile body **11** to a place where the user is.

[0331] In step S**203**, the autonomous mobile body **11** performs action place learning processing.

[0332] Here, details of action place learning processing will be described with reference to a flowchart of FIG. **22**.

[0333] In step S**221**, the autonomous mobile body **11** performs clustering of indicated action places. Specifically, the recognition unit **121** supplies data indicating an action detail and an indicated action place (hereinafter referred to as an indicated place) to the learning unit **122**.

[0334] The learning unit **122** adds data including the action detail and the indicated place of this time to the action history data stored in the storage unit **106**.

[0335] Furthermore, the learning unit **122** reads, from the action history data, the latest N data (including the data of this time) among the data of the action detail similar to the action detail of

this time.
[0336] Then, the learning unit **122** performs clustering of the indicated places indicated in the read data by using, for example, a clustering method such as the K-means method.
[0337] FIG. **23** illustrates a specific example of clustering of the indicated places. In this example, an indicated place Pal to an indicated place Pa**3** are classified into a cluster Pb**1**, and an indicated place Pa**4** to an indicated place Pa**8** are classified into a cluster Pb**2**. In this way, the indicated place Pal to the indicated place Pa**3** are grouped into one action place Pb**1**, and the indicated place Pa**4** to the indicated place Pa**8** are grouped into one action place Pb**2**.
[0338] In step S**222**, the learning unit **122** calculates a degree of certainty of each action place. For example, the learning unit **122** calculates the degree of certainty of each action place on the basis of the number of indicated places included in each action place. For example, in the example of FIG. **22**, since the action place Pb**2** contains more indicated places than the action place Pba, the degree of certainty is set higher.
[0339] The learning unit **122** updates action place data that is one of the learning result data and stored in the storage unit **106**, on the basis of the learning result of this time.
[0340] Note that, the action place data includes, for example, each action detail, and the action place and degree of certainty corresponding to each action detail.
[0341] As described above, the action place where the predetermined action is performed and the degree of certainty for each action place are set on the basis of a distribution of the indicated place in the past.
[0342] Thereafter, the action place learning processing ends.
[0343] Returning to FIG. **21**, on the other hand, in a case where it is determined in step S**202** that the place to act is not indicated, the processing of step S**203** is skipped and the processing proceeds to step S**204**.
[0344] In step S**204**, the action planning unit **123** sets the action place.
[0345] For example, in a case where the place to act is indicated by the user, the action planning unit **123** determines the indicated place as the action place.
[0346] On the other hand, in a case where the place to act is not indicated by the user, the action planning unit **123** determines the action place on the basis of the current position of the autonomous mobile body **11** and the action place data stored in the storage unit **106**.
[0347] Specifically, for example, in a case where the action place corresponding to the action detail of this time is not learned and no action place is set, the action planning unit **123** determines the current position of the autonomous mobile body **11** as the action place.
[0348] On the other hand, in a case where the action place corresponding to the action detail of this time is learned and one or more action places are set, the action planning unit **123** determines the action place on the basis of the degree of certainty of each action place and a distance to each action place. For example, in a case where the action place is set within a range of a predetermined distance, the action planning unit **123** selects the action place with the highest degree of certainty among the action places within the range of the predetermined distance. Note that, in a case where there is only one action place within the range of the predetermined distance, the one action place is selected. On the other hand, for example, in a case where the action place is not set within the range of the predetermined distance, the action planning unit **123** determines the current position of the autonomous mobile body **11** as the action place.
[0349] The action planning unit **123** supplies data indicating the action detail and the action place to the operation control unit **124**.
[0350] In step S**205**, the autonomous mobile body **11** moves to the determined action place. Specifically, the operation control unit **124** controls the drive unit **104** to move the autonomous mobile body **11** to the determined action place.
[0351] In step S**206**, the autonomous mobile body **11** executes the action. Specifically, the operation control unit **124** controls the drive unit **104** and the output unit **105** to cause the

autonomous mobile body **11** to execute the action set by the action planning unit **123**.
[0352] Thereafter, the processing returns to step S**201**, and the processing after step S**201** is executed.
[0353] Here, a specific example of the action control processing of FIG. **21** will be described with reference to FIG. **24**.
[0354] FIG. **24** is a graph illustrating a time series change of the urinary intention coefficient that defines a timing of urination by the autonomous mobile body **11**.
[0355] At time to, when the power of the autonomous mobile body **11** is turned on and the autonomous mobile body **11** is activated, for example, the urinary intention coefficient is set to 0.6.
[0356] Note that, the urinary intention coefficient increases linearly at a predetermined rate (for example, +0.1 in 1 hour) until the autonomous mobile body **11** performs a predetermined action (for example, urinating, eating food, transitioning to the standby mode, or the like) or the urinary intention coefficient reaches 1.0.
[0357] At time t**1**, when the autonomous mobile body **11** performs the operation of eating food, the urinary intention coefficient increases to 0.9. When the urinary intention coefficient is greater than or equal to 0.9, the autonomous mobile body **11** starts an operation indicating that it has a desire to urinate. For example, in a range of the urinary intention coefficient from 0.9 to 0.95, the autonomous mobile body **11** performs a restless motion. Furthermore, for example, in a range of the urinary intention coefficient from 0.95 to 1.0, the restless motion of the autonomous mobile body **11** becomes large.
[0358] When the urinary intention coefficient reaches 1.0 at time t**2**, the autonomous mobile body **11** determines that a condition for urinating is satisfied, and spontaneously urinates at time t**3**. At this time, the place to urinate is determined by the processing of step S**205** of FIG. **21** described above. Furthermore, when urination is performed, the urinary intention coefficient drops to 0.
[0359] At time t**4**, when the autonomous mobile body **11** performs the operation of eating food, the urinary intention coefficient increases to 0.9.
[0360] At time t**5**, for example, in a case where the user instructs the autonomous mobile body **11** to “Pee”, the autonomous mobile body **11** urinates in accordance with the instruction. At this time, the place to urinate is determined by the processing of step S**205** of FIG. **21** described above. Furthermore, when urination is performed, the urinary intention coefficient drops to 0.
[0361] When the autonomous mobile body **11** transitions to the standby mode at time t**6**, a change in the urinary intention coefficient stops, and the urinary intention coefficient is held without increasing or decreasing during the standby mode.
[0362] At time t**7**, when the autonomous mobile body **11** returns from the standby mode, the urinary intention coefficient begins to rise again.
[0363] As described above, the user can perform discipline for the place where the autonomous mobile body **11** performs the predetermined action.
[0364] Note that, by the above-described processing, it is possible to perform discipline for a place where the autonomous mobile body **11** performs any action other than urination.
[0365] For example, it is possible to perform discipline for a place (for example, the entrance or the like) where the autonomous mobile body **11** welcomes the user when the user returns home.
[0366] Furthermore, by similar processing, it is possible to discipline the autonomous mobile body **11** for a place where a predetermined action is prohibited.
[0367] For example, it is possible to discipline the autonomous mobile body **11** for a place where the autonomous mobile body **11** is prohibited from approaching or entering.
[0368] Specifically, for example, in a case where the autonomous mobile body **11** repeatedly hits the same chair, the user repeats an instruction such as “It is not good here because you hits the chair”, “It is not good because it is dangerous here”, or the like when the autonomous mobile body **11** hits the chair, whereby the autonomous mobile body **11** learns the vicinity of the chair as a dangerous place. As a result, the autonomous mobile body **11** comes to move avoiding the vicinity

of the chair gradually.
[0369] Furthermore, for example, the kitchen is a place where the user does not want the autonomous mobile body **11** to enter because the user performs cooking or washing, the floor is wet, and it is dangerous. Thus, for example, every time the autonomous mobile body **11** approaches or enters the kitchen, the user gives an instruction such as “Don't come to the kitchen”. As a result, the autonomous mobile body **11** learns that the kitchen is a place where the autonomous mobile body **11** should not enter, and comes to avoid entering the kitchen gradually.
[0370] Note that, for example, it is also possible to discipline the autonomous mobile body **11** by combining the above-described first method and the second method.
[0371] For example, the user who is a housewife indicates the entrance as a place to welcome the user's master, for the autonomous mobile body **11**, and repeatedly gives an instruction to “Welcome master at the entrance” at a similar time zone. As a result, the autonomous mobile body **11** gradually comes to be able to welcome the master at the entrance when it is the time zone.
[0372] For example, the user indicates a place where the autonomous mobile body **11** waits, and repeatedly gives an instruction to “Wait at the usual place” when the user wants the autonomous mobile body **11** to be quiet or when the user does not want the autonomous mobile body **11** to be near. As a result, the autonomous mobile body **11** gradually comes to be able to stand by at the indicated place. Furthermore, by learning a situation in which the instruction is given, the autonomous mobile body **11** comes to be able to voluntarily move to the indicated place and stand by when the user wants the autonomous mobile body **11** to be quiet or when the user does not want the autonomous mobile body **11** to be near.
[0373] For example, in a case where the autonomous mobile body **11** always stops at the same place, the user gives an instruction such as “You can walk because there is nothing” every time the autonomous mobile body **11** stops at that place. As a result, the autonomous mobile body **11** can learn the place as a safe place and gradually comes to be able to continue walking without stopping at the place.
<Third Method of Disciplining Autonomous Mobile Body **11**>
[0374] Next, a third method of disciplining the autonomous mobile body **11** will be described with reference to FIG. **25**.
[0375] In the third method, the autonomous mobile body **11** is disciplined to execute an action corresponding to an object when the autonomous mobile body **11** remembers the object that triggers execution of the action, that is, the object that triggers the action (hereinafter referred to as an action trigger object) and finds the object, or the like.
<Third Embodiment of Action Control Processing>
[0376] Here, a third embodiment of the action control processing executed by the autonomous mobile body **11** will be described with reference to a flowchart of FIG. **25**.
[0377] The processing is started when the power of the autonomous mobile body **11** is turned on, and ends when the power of the autonomous mobile body **11** is turned off.
[0378] In step S**301**, the recognition unit **121** determines whether or not the object is indicated, on the basis of the sensor data supplied from the input unit **101**. In a case where it is determined that the object is not indicated, the processing proceeds to step S**302**.
[0379] In step S**302**, the recognition unit **121** determines whether or not the object that triggers the action is recognized, on the basis of the sensor data supplied from the input unit **101**. In a case where a new object is not recognized, the recognition unit **121** determines that the object that triggers the action is not recognized, and the processing returns to step S**301**.
[0380] On the other hand, in a case where a new object is recognized, the recognition unit **121** recognizes a type of the object. Furthermore, in the processing of step S**313** described later, the recognition unit **121** reads, from the storage unit **106**, action object correlation data obtained by learning a correlation between each action and each object.
[0381] The action object correlation data is one of the learning result data, and is data indicating a

correlation between each action and an action trigger object that triggers execution of each action. For example, the action object correlation data indicates a correlation between an object indicated by the user and an action for which the instruction is given together with the indication of the object.

[0382] In a case where there is no action trigger object that matches the type of the recognized object in the action object correlation data, the recognition unit **121** determines that the object that triggers the action is not recognized, and the processing returns to step S**301**.

[0383] Thereafter, the processing of steps S**301** and S**302** is repeatedly executed until it is determined in step S**301** that the object is indicated, or it is determined in step S**302** that the object that triggers the action is recognized.

[0384] On the other hand, in step S**301**, for example, in a case where it is recognized that the user points to an object to be remembered by the autonomous mobile body **11** or presents the object in front of the autonomous mobile body **11**, the recognition unit **121** determines that the object is indicated, and the processing proceeds to step S**303**.

[0385] In step S**303**, it is determined whether or not an instruction is given from the user, similarly to the processing of step S**103** of FIG. **17**. In a case where it is determined that the instruction is given from the user, the processing proceeds to step S**304**.

[0386] In step S**304**, the action planning unit **123** sets an action detail on the basis of the user's instruction. Specifically, the recognition unit **121** supplies data indicating the type of the indicated object and the recognized detail of the user's instruction to the action planning unit **123**. The action planning unit **123** sets the action detail on the basis of the type of the indicated object and the detail for which the instruction is given together with the indication of the object. The action planning unit **123** supplies data indicating the set action detail to the operation control unit **124**. Furthermore, the recognition unit **121** supplies data indicating the type of the indicated object to the learning unit **122**.

[0387] Thereafter, the processing proceeds to step S**309**.

[0388] On the other hand, in a case where it is determined in step S**303** that no instruction is given from the user, the processing proceeds to step S**305**.

[0389] In step S**305**, the recognition unit **121** determines whether or not the object that triggers the action is indicated. Specifically, in a case where there is an action trigger object that matches the type of the indicated object in the action object correlation data, the recognition unit **121** determines that the object that triggers the action is indicated, and the processing proceeds to step S**306**.

[0390] In step S**306**, the action planning unit **123** sets the action detail on the basis of the indicated object. Specifically, the recognition unit **121** supplies data indicating a type of the indicated action trigger object to the learning unit **122** and the action planning unit **123**.

[0391] The action planning unit **123** sets the action detail on the basis of the type of the indicated action trigger object and the action object correlation data stored in the storage unit **106**. That is, an action detail associated with the indicated object (action trigger object) in the action situation correlation data is set as the action detail to be executed by the autonomous mobile body **11**. The action planning unit **123** supplies data indicating the set action detail to the operation control unit **124**.

[0392] Thereafter, the processing proceeds to step S**309**.

[0393] On the other hand, in step S**305**, in a case where there is no action trigger object that matches the type of the indicated object in the action object correlation data, the recognition unit **121** determines that the object that triggers the action is not indicated, and the processing proceeds to step S**307**.

[0394] In step S**307**, the action planning unit **123** sets a predetermined action detail. Specifically, the recognition unit **121** supplies data indicating that an object different from the action trigger object (hereinafter referred to as a non-action trigger object) is indicated and a type of the non-

action trigger object to the action planning unit **123**. Furthermore, the recognition unit **121** supplies data indicating the type of the non-action trigger object to the learning unit **122**.
[0395] Here, for example, an action detail in a case where the non-action trigger object is indicated is set in advance. The action planning unit **123** supplies data indicating the action detail to the operation control unit **124**.
[0396] Thereafter, the processing proceeds to step S**309**.
[0397] On the other hand, in step S**302**, in a case where there is an action trigger object that matches the type of the recognized object in the action object correlation data, the recognition unit **121** determines that the object that triggers the action is recognized, and the processing proceeds to step S**308**.
[0398] In step S**308**, the action detail is set on the basis of the recognized object by processing similar to step S**306**. Furthermore, the recognition unit **121** supplies data indicating a type of the recognized action trigger object to the learning unit **122**.
[0399] Thereafter, the processing proceeds to step S**309**.
[0400] In step S**309**, the action is started, similarly to the processing of step S**106** of FIG. **17**.
[0401] In step S**310**, it is determined whether or not the action is completed, similarly to the processing of step S**107** of FIG. **17**. In a case where it is determined that the action is not completed, the processing proceeds to step S**311**.
[0402] In step S**311** it is determined whether or not to stop the action, similarly to the processing of step S**108** of FIG. **17**. In a case where it is determined not to stop the action, the processing returns to step S**310**.
[0403] Thereafter, the processing of steps S**310** and S**311** is repeatedly executed until it is determined in step S**310** that the action is completed, or it is determined in step S**311** to stop the action.
[0404] On the other hand, in a case where it is determined in step S**310** that the action is completed, the processing proceeds to step S**313**.
[0405] Furthermore, in a case where it is determined in step S**311** to stop the action, the processing proceeds to step S**312**.
[0406] In step S**312**, the action is stopped, similarly to the processing of step S**110** of FIG. **17**.
[0407] Thereafter, the processing proceeds to step S**313**.
[0408] In step S**313**, the learning unit **122** learns a correlation between the action and the object.
[0409] Specifically, the action planning unit **123** supplies the executed action detail and data indicating whether or not the action is completed to the learning unit **122**.
[0410] The learning unit **122** adds a result of the action of this time to the action history data stored in the storage unit **106**.
[0411] Furthermore, the learning unit **122** updates the action object correlation data by learning the correlation between each action and the object on the basis of the action history data. For example, the learning unit **122** calculates a degree of correlation (for example, a correlation coefficient) between each action and each object.
[0412] Note that, for example, in a case where a certain action is completed, the degree of correlation between the action and an object that triggers the action is made large. On the other hand, for example, in a case where a certain action is stopped in the middle, the degree of correlation between the action and an object that triggers the action is made small.
[0413] Then, the learning unit **122** associates an object having a degree of correlation greater than or equal to a predetermined threshold value as an action trigger object, for each action.
[0414] Thus, for example, in a case where the user repeatedly presents the same type of object and gives a similar instruction in the past, the object is associated with the action corresponding to the instruction, as an action trigger object.
[0415] Thereafter, the processing returns to step S**301**, and the processing after step S**301** is executed.

[0416] As described above, it is possible to discipline the autonomous mobile body **11** for the action using the object.

[0417] For example, the user repeatedly presents the user's favorite object to the autonomous mobile body **11** and speaks to the autonomous mobile body **11**, whereby the autonomous mobile body **11** gradually comes to be able to perform an action of being interested in the object. For example, a child repeatedly presents the child's favorite toy to the autonomous mobile body **11** and speaks "Let's play", whereby the autonomous mobile body **11** gradually comes to play with the toy. Furthermore, for example, in a case where the autonomous mobile body **11** finds the toy, the autonomous mobile body **11** voluntarily plays with the toy.

[0418] For example, the user repeatedly presents a food toy to the autonomous mobile body **11** and gives instructions such as "Eat", "Food", and "Remember food", whereby the autonomous mobile body **11** gradually comes to recognize the toy as food. Then, for example, the autonomous mobile body **11** gradually comes to perform an action of eating the toy in a case where the toy is presented. Furthermore, for example, in a case where the autonomous mobile body **11** finds the toy, the autonomous mobile body **11** comes to voluntarily perform the action of eating the toy.

[0419] On the contrary, for example, the user repeatedly presents a toy that is not food to the autonomous mobile body **11** and gives instructions such as "You can't eat it" and "It is not food", whereby the autonomous mobile body **11** gradually comes to recognize that the toy is not food. Then, for example, the autonomous mobile body **11** gradually comes to stop performing the action of eating the toy.

[0420] Note that, for example, it is also possible to discipline the autonomous mobile body **11** by combining the above-described second method and the third method.

[0421] For example, the user repeatedly moves the autonomous mobile body **11** to a place where the toy is to be put away, presents the toy, and gives an instruction such as "Put it away here". As a result, the autonomous mobile body **11** gradually remembers the place and action to put the toy away. For example, at the beginning, the autonomous mobile body **11** sometimes does not put the toy away or puts the toy away in a wrong place even if the user gives an instruction "Put the toy away". However, in the end, the autonomous mobile body **11** comes to be able to put the toy away in an appropriate place if the user gives the instruction "Put the toy away". Furthermore, for example, the autonomous mobile body **11** comes to be able to voluntarily put the toy away in the appropriate place when finding the toy.

<Example of User Interface of Information Processing Terminal **12**>

[0422] Next, an example of the user interface of the information processing terminal **12** regarding the discipline for the autonomous mobile body **11** will be described with reference to FIGS. **26** to **35**.

[0423] FIGS. **26** to **28** illustrate display examples of a display **501** of the information processing terminal **12** in a case where the autonomous mobile body **11** is executing a silent action.

[0424] FIG. **26** illustrates an example of a home screen. Icons **511** to **513** and an animation **514** are displayed on the home screen.

[0425] In a case where the icon **511** in the upper left corner of the home screen is pressed, a menu screen is displayed. By transitioning from the menu screen to another screen, the user can confirm, for example, advice on how to raise the autonomous mobile body **11**, information regarding the user, and the like.

[0426] When the icon **512** in the upper right corner of the home screen is pressed, an information screen is displayed. On the information screen, for example, information regarding the autonomous mobile body **11**, information from a company that provides a service regarding the autonomous mobile body **11**, and the like are displayed. Furthermore, the number of pieces of unread information is displayed in the upper left of the icon **512**. Note that, a specific example of the information screen will be described later.

[0427] When the icon **513** at the center and bottom of the home screen is pressed, an information confirmation screen is displayed. On the information confirmation screen, for example, information regarding a character, gender, action, remaining battery level, and the like of the autonomous mobile body **11** is displayed.

[0428] The animation **514** is an animation that imitates the autonomous mobile body **11**, and represents a current state of the autonomous mobile body **11**. In this example, a cross-marked image **521** indicating that the autonomous mobile body **11** is in the silent action is displayed in a mouth portion of the animation **514**.

[0429] FIG. **27** illustrates an example of the information screen displayed in a case where the icon **512** on the home screen of FIG. **26** is pressed.

[0430] Note that, in a "robot" portion of a sentence in the information screen, a product name or a name of the autonomous mobile body **11** is actually entered. This also applies to the following screen examples.

[0431] In the second row from the top of the information screen, a situation of the discipline for the silent action of the autonomous mobile body **11** is indicated. In this example, it is indicated that the autonomous mobile body **11** gradually remembers a thing for which the autonomous mobile body **11** is told "Be quiet", and the discipline is progressed.

[0432] FIG. **28** illustrates an example of a screen displaying details of the information in the second row from the top of the information screen of FIG. **27**. In this example, information is displayed indicating a proficiency level for the discipline for the silent action of the autonomous mobile body **11**. Specifically, it is indicated that the autonomous mobile body **11** becomes more patient and quieter as the autonomous mobile body **11** gets used to it. Furthermore, it is indicated that when the name of the autonomous mobile body **11** is called, the silent action is stopped and motion is started.

[0433] Note that, for example, the proficiency level may be represented by a specific numerical value or the like.

[0434] FIGS. **29** to **31** illustrate display examples of the display **501** of the information processing terminal **12** in a case where the autonomous mobile body **11** is sleeping.

[0435] FIG. **29** illustrates an example of the home screen. Note that, in the figure, portions corresponding to those in FIG. **26** are designated by the same reference numerals, and the description thereof will be omitted as appropriate.

[0436] In the home screen of FIG. **29**, the icons **511** to **513** and the animation **514** are displayed, similarly to the home screen of FIG. **26**. Furthermore, a button **541** and an icon **542** are added.

[0437] The animation **514** indicates that the autonomous mobile body **11** is sleeping.

[0438] The button **541** is displayed, for example, in a case where one user owns a plurality of the autonomous mobile bodies **11**. When the button **541** is pressed, for example, a screen is displayed for selecting the autonomous mobile body **11** for which information is to be confirmed.

[0439] The icon **542** indicates a time when the autonomous mobile body **11** is scheduled to wake up. In this example, it is indicated that the autonomous mobile body **11** is scheduled to wake up around seven o'clock.

[0440] FIG. **30** illustrates an example of the information confirmation screen displayed in a case where the icon **513** on the home screen of FIG. **29** is pressed. In this example, a message is displayed indicating that the autonomous mobile body **11** is sleeping and is scheduled to wake up around seven o'clock.

[0441] Furthermore, a graph is displayed indicating a transition (biorhythm) of an amount of activity of the autonomous mobile body **11**. In this example, it is indicated that the amount of activity of the autonomous mobile body **11** decreases during a period from 22:00 to 7:00.

[0442] FIG. **31** illustrates an example of a setting screen displayed in a case where the icon **511** of the home screen of FIG. **29** is pressed. On the setting screen, various setting items regarding the autonomous mobile body **11** are displayed. Among these setting items, in the time zone, the user can explicitly set a bedtime and a wake-up time of the autonomous mobile body **11**.

[0443] FIGS. **32** to **35** illustrate display examples of the display **501** of the information processing

terminal **12** in a case of confirming the proficiency level and the like of the autonomous mobile body **11** for discipline.
[0444] FIG. **32** illustrates an example of a main screen for confirming the proficiency level of the autonomous mobile body **11** for discipline.
[0445] Graphs **601** to **604** are displayed to be arranged in the vertical direction on the main screen.
[0446] The graph **601** indicates an overall proficiency level of the autonomous mobile body **11**. The overall proficiency level is calculated on the basis of the proficiency level for each discipline item.
[0447] The graph **602** indicates the proficiency level for discipline at a urination place (toilet place).
[0448] The graph **603** illustrates the proficiency level for discipline for the silent action.
[0449] The graph **604** illustrates the proficiency level for discipline for the sleeping action.
[0450] FIG. **33** illustrates an example of a screen displayed in a case where the graph **602** is selected in the main screen of FIG. **32**.
[0451] In this example, a message **621** is displayed to guide how to discipline the urination place.
[0452] Below the message **621**, a graph **622** is displayed indicating the proficiency level for discipline for the urination place.
[0453] Below the graph **622**, a map **623** is displayed indicating the currently set urination place although detailed illustration is omitted. Note that, instead of the map **623**, a link to the map indicating the urination place may be displayed.
[0454] FIG. **34** illustrates an example of a screen displayed in a case where the graph **603** is selected on the main screen of FIG. **32**.
[0455] In this example, a message **641** is displayed to guide how to discipline the silent action.
[0456] Below the message **641**, a graph **642** is displayed indicating the proficiency level for discipline for the silent action.
[0457] FIG. **35** illustrates an example of a screen displayed in a case where the graph **604** is selected in the main screen of FIG. **32**.
[0458] In this example, a message **661** is displayed to guide how to discipline the sleeping action.
[0459] Below the message **661**, a graph **662** is displayed indicating the proficiency level for discipline for the sleeping action.
[0460] Below the graph **662**, for example, a graph is displayed similar to the graph of the amount of activity of FIG. **30** although detailed illustration is omitted.
[0461] As described above, the user can confirm the current state of the autonomous mobile body **11**, how to discipline, the proficiency level for discipline, and the like by using the information processing terminal **12**.
2. Modifications
[0462] Hereinafter, modifications of the above-described embodiment of the present technology will be described.
[0463] For example, by being disciplined for a certain action, the autonomous mobile body **11** may naturally improve the proficiency level for another action without being disciplined for the other action. In particular, the proficiency may be made to improve more in an action that is similar to or related to an action for which the proficiency level is improved due to discipline.
[0464] For example, in a case where an interval at which the user performs discipline is widened to greater than or equal to a predetermined threshold value, the autonomous mobile body **11** may gradually forget an action corresponding to the discipline. For example, in a case where the user does not perform discipline for a certain action for a long period of time, the proficiency level for the action may be decreased. Note that, for example, in a case where the proficiency level for a certain action exceeds a predetermined threshold value, the proficiency level for the action may be made not to decrease even if discipline for the action is not performed for a long period of time.
[0465] For example, the autonomous mobile body **11** may select a person from whom the

autonomous mobile body **11** is disciplined. For example, in a case where a person recognized by the autonomous mobile body **11** is the user or the user's family or relative, the autonomous mobile body **11** may be disciplined by the person, and in a case where the recognized person is another person, the autonomous mobile body **11** may be made not to be disciplined by the other person.

[0466] For example, the number of times of instruction action described above may include the number of times the action for which the instruction is given is stopped in the middle and not completed.

3. Others

<Configuration Example of Computer>

[0467] A series of processing steps described above can be executed by hardware, or can be executed by software. In a case where the series of processing steps is executed by software, a program constituting the software is installed in a computer. Here, the computer includes a computer incorporated in dedicated hardware, and a computer capable of executing various functions by installation of various programs, for example, a general purpose personal computer, and the like.

[0468] FIG. **36** is a block diagram illustrating a configuration example of hardware of the computer that executes the above-described series of processing steps by the program.

[0469] In a computer **1000**, a central processing unit (CPU) **1001**, a read only memory (ROM) **1002**, and a random access memory (RAM) **1003** are connected to each other by a bus **1004**.

[0470] Moreover, an input/output interface **1005** is connected to the bus **1004**. The input/output interface **1005** is connected to an input unit **1006**, an output unit **1007**, a recording unit **1008**, a communication unit **1009**, and a drive **1010**.

[0471] The input unit **1006** includes an input switch, a button, a microphone, an imaging element, and the like. The output unit **1007** includes a display, a speaker, and the like. The recording unit **1008** includes a hard disk, a nonvolatile memory, or the like. The communication unit **1009** includes a network interface and the like. The drive **1010** drives a removable medium **1011** such as a magnetic disk, an optical disk, a magneto-optical disk, or a semiconductor memory.

[0472] In the computer **1000** configured as described above, for example, the CPU **1001** loads the program recorded in the recording unit **1008** to the RAM **1003** via the input/output interface **1005** and the bus **1004** to execute the above-described series of processing steps.

[0473] The program executed by the computer **1000** (CPU **1001**) can be provided, for example, by being recorded in the removable medium **1011** as a package medium or the like. Furthermore, the program can be provided via a wired or wireless transmission medium such as a local area network, the Internet, or digital satellite broadcasting.

[0474] In the computer **1000**, the program can be installed to the recording unit **1008** via the input/output interface **1005** by mounting the removable medium **1011** to the drive **1010**. Furthermore, the program can be received by the communication unit **1009** via the wired or wireless transmission medium, and installed to the recording unit **1008**. In addition, the program can be installed in advance to the ROM **1002** or the recording unit **1008**.

[0475] Note that, the program executed by the computer can be a program by which the processing is performed in time series along the order described in the present specification, and can be a program by which the processing is performed in parallel or at necessary timing such as in a case where a call is performed.

[0476] Furthermore, in the present specification, a system means an aggregation of a plurality of constituents (device, module (component), and the like), and it does not matter whether or not all of the constituents are in the same cabinet. Thus, a plurality of devices that is accommodated in a separate cabinet and connected to each other via a network and one device that accommodates a plurality of modules in one cabinet are both systems.

[0477] Moreover, the embodiment of the present technology is not limited to the embodiment described above, and various modifications are possible without departing from the scope of the

present technology.
[0478] For example, the present technology can adopt a configuration of cloud computing that shares one function in a plurality of devices via a network to perform processing in cooperation.
[0479] Furthermore, each step described in the above flowchart can be executed by sharing in a plurality of devices, other than being executed by one device.
[0480] Moreover, in a case where a plurality of pieces of processing is included in one step, the plurality of pieces of processing included in the one step can be executed by sharing in a plurality of devices, other than being executed by one device.
<Combination Example of Configurations>
[0481] The present technology can also have a configuration as follows.
[0482] (1)
[0483] An autonomous mobile body including: [0484] a recognition unit that recognizes an instruction given; [0485] an action planning unit that plans an action on the basis of the instruction recognized; and [0486] an operation control unit that controls execution of the action planned, in which [0487] the action planning unit changes a detail of a predetermined action as an action instruction that is an instruction for the predetermined action is repeated.
[0488] (2)
[0489] The autonomous mobile body according to (1), in which [0490] the action planning unit changes the detail of the predetermined action on the basis of the number of times of the instruction that is the number of times the action instruction is given.
[0491] (3)
[0492] The autonomous mobile body according to (2), in which [0493] the action planning unit improves a proficiency level for the predetermined action as the number of times of the instruction increases.
[0494] (4)
[0495] The autonomous mobile body according to (3), in which [0496] the action planning unit improves the proficiency level for the predetermined action as the number of times the predetermined action is completed increases out of the number of times of the instruction.
[0497] (5)
[0498] The autonomous mobile body according to (3) or (4), in which [0499] the action planning unit improves the proficiency level for the predetermined action and improves a proficiency level for an action different from the predetermined action.
[0500] (6)
[0501] The autonomous mobile body according to any of (3) to (5), in which [0502] the action planning unit lowers the proficiency level for the predetermined action in a case where an interval of the action instruction is widened.
[0503] (7)
[0504] The autonomous mobile body according to any of (1) to (6), in which [0505] the recognition unit further recognizes a situation, [0506] a learning unit is further included, the learning unit learning a correlation between the situation in which the instruction is given and the action for which the instruction is given, and [0507] the action planning unit plans the action on the basis of the situation recognized by the recognition unit and a result of learning the correlation.
[0508] (8)
[0509] The autonomous mobile body according to (7), in which [0510] in a case where the action for which the instruction is given is completed, the learning unit increases a degree of correlation between the situation in which the instruction is given and the action for which the instruction is given, and in a case where the action for which the instruction is given is stopped in the middle, the learning unit decreases the degree of correlation.
[0511] (9)
[0512] The autonomous mobile body according to (7) or (8), in which [0513] the learning unit

estimates a time series transition of an expected value for the predetermined action on the basis of a distribution of a time at which the instruction for the predetermined action is given, and [0514] the action planning unit plans the action on the basis of the expected value.
[0515] (10)
[0516] The autonomous mobile body according to (9), in which [0517] the action planning unit increases the expected value in a case where an instruction to promote the predetermined action is given, and decreases the expected value in a case where an instruction to suppress the predetermined action is given.
[0518] (11)
[0519] The autonomous mobile body according to (9) or (10), in which [0520] the action planning unit plans the predetermined action or an action similar to the predetermined action in a case where the expected value is greater than or equal to a predetermined threshold value.
[0521] (12)
[0522] The autonomous mobile body according to any of (1) to (11), further including [0523] a learning unit that learns an action place that is a place where the predetermined action is performed, on the basis of a history of an indicated place that is a place for which an instruction to perform the predetermined action is given in the past, in which [0524] the action planning unit determines the place where the predetermined action is performed, on the basis of a result of learning the action place.
[0525] (13)
[0526] The autonomous mobile body according to (12), in which [0527] the learning unit sets one or more of the action places on the basis of a distribution of the indicated place, and [0528] the action planning unit determines the place where the predetermined action is performed, on the basis of the action place and a current position.
[0529] (14)
[0530] The autonomous mobile body according to (13), in which [0531] the learning unit sets a degree of certainty of the action place on the basis of the distribution of the indicated place, and [0532] the action planning unit determines the place where the predetermined action is performed, on the basis of the action place, the degree of certainty, and the current position.
[0533] (15)
[0534] The autonomous mobile body according to (13) or (14), in which [0535] the learning unit sets the action place by performing clustering of the indicated place.
[0536] (16)
[0537] The autonomous mobile body according to any of (1) to (15), in which [0538] the recognition unit further recognizes an object, [0539] a learning unit is further included, the learning unit learning a correlation between the object indicated and an action for which an instruction is given along with indication of the object, and [0540] the action planning unit plans the action on the basis of the object recognized by the recognition unit and a result of learning the correlation.
[0541] (17)
[0542] An information processing method including: [0543] recognizing an instruction given to an autonomous mobile body; [0544] planning an action of the autonomous mobile body on the basis of the instruction recognized, and changing a detail of a predetermined action as an action instruction that is an instruction for the predetermined action is repeated; and [0545] controlling execution of the action of the autonomous mobile body planned.
[0546] (18)
[0547] A program for causing a computer to execute processing of: [0548] recognizing an instruction given to an autonomous mobile body; [0549] planning an action of the autonomous mobile body on the basis of the instruction recognized, and changing a detail of a predetermined action as an action instruction that is an instruction for the predetermined action is repeated; and [0550] controlling execution of the action of the autonomous mobile body planned.

[0551] (19)
[0552] An information processing device including: [0553] a recognition unit that recognizes an instruction given to an autonomous mobile body; [0554] an action planning unit that plans an action of the autonomous mobile body on the basis of the instruction recognized; and [0555] an operation control unit that controls execution of the action of the autonomous mobile body planned, in which [0556] the action planning unit changes a detail of a predetermined action by the autonomous mobile body as an action instruction that is an instruction for the predetermined action is repeated.
[0557] Note that, the advantageous effects described in the present specification are merely examples, and the advantageous effects of the present technology are not limited to them and may include other effects.
REFERENCE SIGNS LIST
[0558] **1** Information processing system [0559] **11** Autonomous mobile body [0560] **12** Information processing terminal [0561] **1** Information processing server [0562] **103** Information processing unit [0563] **104** Drive unit [0564] **105** Output unit [0565] **121** Recognition unit [0566] **122** Learning unit [0567] **123** Action planning unit [0568] **124** Operation control unit [0569] **151** Expected value calculation unit [0570] **203** Information processing unit [0571] **204** Output unit [0572] **302** Information processing unit [0573] **321** Autonomous mobile body control unit [0574] **322** Application control unit [0575] **331** Recognition unit [0576] **332** Learning unit [0577] **333** Action planning unit [0578] **334** Operation control unit

Claims

1. An information processing apparatus, comprising: circuitry configured to: recognize at least one of a current situation or an object that is currently present around an autonomous mobile body based on at least one of sensor data or information acquired via a network; determine the recognized at least one of situation or the object currently present around the autonomous mobile body is similar to a stored situation or a stored object that is associated with a user utterance; and control the autonomous mobile body to initiate an action associated with the user utterance based on the determination that the recognized at least one of the current situation or the object currently present around the autonomous mobile body is similar to the stored situation or the stored object that is associated with the user utterance.
2. The information processing apparatus according to claim 1, wherein the circuitry is further configured to determine whether a first instruction is received from a user.
3. The information processing apparatus according to claim 2, wherein the circuitry is further configured to recognize the current situation based on the determination that the first instruction is received from the user.
4. The information processing apparatus according to claim 2, wherein the circuitry is further configured to determine whether the current situation requires initiation of the action based on the determination that the first instruction is not received from the user.
5. The information processing apparatus according to claim 4, wherein the circuitry is further configured to set an action detail based on the current situation.
6. The information processing apparatus according to claim 3, wherein the circuitry is further configured to set an action detail based on a number of times the action associated with the first instruction has been taken in past.
7. The information processing apparatus according to claim 6, wherein the circuitry is further configured to: determine whether the action has been completed; and learn a correlation between the action and the current situation based on the completion of the action.
8. The information processing apparatus according to claim 7, wherein the circuitry is further configured to: determine whether the action is to be stopped before completion based on a second

instruction that triggers to stop the action; and stop the action based on the determination that the action is to be stopped before the completion.

9. The information processing apparatus according to claim 1, wherein the circuitry is further configured to determine whether a condition for execution of the action is satisfied based on the at least one of the sensor data or the information acquired via the network.

10. The information processing apparatus according to claim 9, wherein the circuitry is further configured to determine whether a place to act is indicated based on the determination that the condition for the execution of the action is satisfied.

11. The information processing apparatus according to claim 10, wherein the circuitry is further configured to execute an action place learning processing based on the determination that the place to act is indicated.

12. The information processing apparatus according to claim 1, wherein the circuitry is further configured to determine an action place for the action to be executed by the autonomous mobile body based on one of: a user input, or a current position of the autonomous mobile body and pre-stored action place data.

13. The information processing apparatus according to claim 1, wherein the circuitry is further configured to: control the autonomous mobile body to move toward the determined action place; and control the autonomous mobile body to execute the action at the action place.

14. A system, comprising: an autonomous mobile body; and an information processing apparatus that includes circuitry, wherein the circuitry is configured to: recognize at least one of a current situation or an object that is currently present around the autonomous mobile body based on at least one of sensor data or information acquired via a network; determine the recognized at least one of situation or the object currently present around the autonomous mobile body is similar to a stored situation or a stored object that is associated with a user utterance; and control the autonomous mobile body to initiate an action associated with the user utterance based on the determination that the recognized at least one of the current situation or the object currently present around the autonomous mobile body is similar to the stored situation or the stored object that is associated with the user utterance.

15. A non-transitory computer-readable medium having stored thereon, computer executable instructions, which when executed by a computer, cause the computer to execute operations, the operations comprising: recognizing at least one of a current situation or an object that is currently present around an autonomous mobile body based on at least one of sensor data or information acquired via a network; determining the recognized at least one of situation or the object currently present around the autonomous mobile body is similar to a stored situation or a stored object that is associated with a user utterance; and controlling the autonomous mobile body to initiate an action associated with the user utterance based on the determination that the recognized at least one of the current situation or the object currently present around the autonomous mobile body is similar to the stored situation or the stored object that is associated with the user utterance.